

AI-POWERED AUTONOMOUS PERSONAL SAFETY ECOSYSTEM

SENTRIX CORE

Architecture, Multi-Modal Sensor Fusion, and a Comparative Review of the
Wearable Personal-Safety Technology Landscape

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Submitted in partial fulfillment of the requirements of the graduation project / research program

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SCOPE OF THIS PAPER

A thirteen-engine AI decision architecture (E1-E13) unifying multi-modal sensor fusion, a personalized behavioral Digital Twin, edge-cloud hybrid inference, and predictive risk scoring; a companion mobile application specified as a seventy-four-screen taxonomy; and an EVT-grade smart-band hardware specification (MCU, sensors, connectivity, power, PCB, and mechanical design). Benchmarked against ten commercial wearable-safety systems and supported by twenty public datasets and over eighty academic, industrial, and dataset references in IEEE citation format.

Keywords: Wearable AI · Personal Safety · Fall Detection · Digital Twin · Sensor Fusion · Edge AI · Mobile Application Architecture

DOCUMENT TYPE

Research Paper / Graduation Project Report / Technical White Paper

VERSION

Publisher Edition v2.0

STATUS

Final Draft — For Review

DATE ISSUED

July 2026

CONFIDENTIAL — For Academic and Technical Review Purposes Only

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Abstract

Wearable technologies have advanced rapidly over the past decade, enabling continuous physiological, behavioral, and environmental sensing through consumer-grade devices. However, the majority of current commercial and research systems remain reactive: they respond to an adverse event only after it has occurred, rely on population-level models rather than individualized baselines, and treat health monitoring and personal safety as separate product categories. This paper presents SENTRIX CORE, a proposed AI-powered autonomous personal safety ecosystem that unifies multi-modal sensor fusion, a personalized behavioral Digital Twin, edge-cloud hybrid inference, and predictive risk scoring into a single decision architecture capable of autonomous emergency escalation. We review the academic literature on fall detection, human activity recognition, sensor fusion, Digital Twin healthcare, and edge AI, and we compare SENTRIX CORE against the current commercial landscape, including Apple Watch, Fitbit, Garmin, WHOOP, Oura, Medical Guardian, AngelSense, and Empatica. Based on this review, we identify five recurring gaps in existing systems and describe how a thirteen-engine AI architecture, together with a curated set of public benchmark datasets, could address them. We further specify the companion mobile application that exposes this architecture to end users - a seventy-four-screen taxonomy spanning onboarding, daily health and location monitoring, specialized modes for children, elderly users, and women's safety, a complete emergency and incident-review subsystem, and a Guardian caregiving layer — together with its Phone-Only and Full Wearable operating modes and a wider multi-platform companion ecosystem, and we specify the physical wearable itself as an Engineering Validation Test (EVT) prototype: a nine-subsystem, MCU-orchestrated hardware platform with a defined sensor suite, communication subsystem, power system, bill of materials, PCB layout, mechanical design, and firmware architecture. We conclude with an assessment of patentable sub-systems, a research novelty statement, and a state-of-the-art comparison table.

Keywords: Wearable AI, Personal Safety, Fall Detection, Digital Twin, Sensor Fusion, Edge AI, Predictive Health Monitoring, TinyML, Human Activity Recognition, Emergency Response Systems

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1. Problem Statement

1.1 Background

The rapid development of wearable technologies has created new opportunities for continuous human monitoring, healthcare management, and personal safety enhancement. Modern wearable devices such as smartwatches and dedicated medical monitoring systems are capable of collecting physiological, behavioral, and environmental information through embedded sensors, including accelerometers, gyroscopes, optical heart-rate sensors, GPS modules, temperature sensors, and microphones.

These devices have demonstrated significant value in fitness monitoring, chronic disease management, elderly care, and emergency detection. However, current wearable ecosystems remain limited because they are primarily designed as monitoring devices rather than intelligent, autonomous safety systems.

1.2 Current Limitations

Problem 1 — Reactive Safety Instead of Predictive Intelligence

Most existing safety wearables operate only after an adverse event occurs. A conventional fall detection pipeline follows a strictly sequential pattern: a fall occurs, an impact is detected, and an alert is generated. This reactive approach cannot answer whether a user is becoming unstable, whether a fall is likely to happen, whether a user's physiological state is deteriorating, or whether the user is entering a dangerous situation. Prior research on wearable fall detection has shown that traditional event-detection approaches continue to suffer from false positives, false negatives, and poor personalization [1], [2].

Problem 2 — Lack of Personalized Intelligence

Most wearable systems rely on generalized models trained on population-level datasets, yet human physiology and behavior differ significantly between individuals. A heart rate of 100 BPM may be entirely normal for an athlete immediately after exercise but abnormal for an elderly person at rest; a given movement pattern may represent normal walking for one person and a sign of instability for another. Future systems therefore require personalized AI models capable of continuously learning user-specific patterns — a concept commonly referred to as a **Human Digital Twin**, a continuously updated computational model of a real person built from sensor data, historical behavior, physiological measurements, and AI prediction models [3], [4].

Problem 3 — Fragmentation Between Health Monitoring and Personal Safety

Current solutions are split across two disconnected product categories. Healthcare wearables such as Apple Watch, Fitbit, and Garmin focus on heart rate, fitness, sleep, and ECG. Personal safety devices such as Medical Guardian and AngelSense focus on SOS alerting, GPS tracking, and emergency communication. No unified intelligent system currently combines health state, physical activity, location, behavior, environment, and emergency response into a single AI decision layer.

Problem 4 — Smartphone Dependency

Many wearable safety solutions depend on a Bluetooth connection, smartphone availability, and internet connectivity. This creates practical limitations for elderly users, children, outdoor workers, and emergency situations in which a paired phone may be absent, discharged, or deliberately separated from the user. A safety device intended for critical situations should be capable of operating independently through cellular (LTE) communication, embedded AI inference, and local decision-making.

Problem 5 — Limited Multi-Modal AI Fusion

Human safety cannot be reliably determined from a single signal. A fall event, for example, should not be inferred from acceleration alone. A robust decision requires combining motion information, heart-rate changes, location context, the acoustic environment, and the user's own history. Research on sensor fusion has repeatedly shown that combining modalities improves robustness relative to single-sensor approaches [5], [6].

1.3 Research Objectives

The objective of the SENTRIX CORE research program is to design an AI-powered autonomous safety ecosystem capable of the following:

- Continuous, multi-modal monitoring of the user's physiological and behavioral state.
- Fusion of heterogeneous sensor streams (motion, cardiac, audio, location) into a unified risk model.
- Generation of a personalized behavioral and physiological Digital Twin for each user.
- Real-time, predictive (not merely reactive) risk assessment.
- Autonomous emergency decision-making with configurable escalation logic.
- Guardian-based monitoring for dependents such as children, elderly relatives, or at-risk individuals.
- Edge AI inference that functions without dependence on a paired smartphone.

2. Literature Review

This section surveys five overlapping bodies of research relevant to SENTRIX CORE: consumer wearable healthcare platforms, fall detection algorithms, edge AI for embedded inference, Digital Twin healthcare modeling, and remote patient monitoring platforms.

2.1 Wearable Healthcare Systems

Apple Watch Healthcare Platform

Apple Watch introduced several consumer healthcare capabilities, including ECG measurement, irregular heart-rhythm notifications, fall detection, and Emergency SOS [7]. The platform benefits from large-scale deployment, strong consumer acceptance, and clinical validation studies such as the Apple Heart Study conducted with Stanford Medicine [52]. Its main limitations are a closed ecosystem, limited customization, and the absence of a complete personal-safety intelligence layer combining prediction and personalization.

Fitbit Healthcare Research

Fitbit devices provide heart-rate monitoring, sleep analysis, and activity recognition, and Fitbit data has been used in multiple independent studies of cardiovascular monitoring and physical-activity estimation [8]. The platform is, however, mainly wellness-focused, with limited autonomous emergency capability.

Garmin Wearable Systems

Garmin devices provide GPS tracking, fitness monitoring, stress estimation, and proprietary “Body Battery” energy metrics [9]. The product line is designed primarily for athletes and outdoor sports, with limited AI-based safety reasoning.

2.2 Fall Detection Research

Fall detection is among the most extensively researched wearable safety applications. Early threshold-based methods used acceleration magnitude, velocity change, and orientation to flag a fall, but these approaches are prone to false alarms and generalize poorly across users [1], [2]. Subsequent machine-learning approaches applied Support Vector Machines and related classifiers to extracted motion features, improving robustness. More recent work applies deep learning architectures — convolutional neural networks (CNNs), long short-term memory networks (LSTMs), and transformer architectures — which automatically learn discriminative features and generally outperform hand-engineered pipelines [10].

2.3 Edge AI Wearables

Edge AI enables local, on-device processing, reducing latency and improving privacy relative to cloud-only inference. Relevant technologies include TinyML, TensorFlow Lite Micro, and low-power ARM Cortex-class microcontrollers [11]. The principal constraints of this approach are limited on-device memory and strict energy budgets, which require careful model compression and quantization.

2.4 Digital Twin Healthcare

Digital Twin research has expanded rapidly in healthcare, with applications in personalized treatment planning, patient simulation, and disease-progression prediction [3], [12], [13]. Existing healthcare Digital Twins, however, are predominantly hospital-oriented and cloud-based rather than real-time, continuously updating wearable systems designed for daily personal use.

2.5 Remote Patient Monitoring

Commercial remote patient monitoring (RPM) platforms such as Biofourmis, Current Health, and Philips RPM combine wearable sensors with cloud analytics and clinician-facing dashboards. These platforms are designed for structured medical-monitoring contexts and healthcare providers rather than for general-purpose consumer safety ecosystems covering children, elderly independence, or outdoor emergencies.

3. Related Work Analysis

This section situates SENTRIX CORE within the existing commercial and academic landscape. A recurring question for any new wearable-safety proposal is whether it meaningfully differs from established consumer products — most notably Apple Watch — or whether it is simply additional functionality layered onto an existing category. Section 3.1 addresses this question directly; Sections 3.2–3.6 then provide the detailed commercial, academic, and comparative-matrix analysis that supports it.

3.1 Market Positioning: Detection Versus Prediction

Apple has built one of the most advanced consumer wearable ecosystems in the world, and any comparative analysis must take this seriously rather than dismiss it. Apple Watch provides heart rate monitoring, ECG, blood-oxygen monitoring (availability varies by region), sleep tracking, respiratory-rate and temperature sensing, fall detection, crash detection, Emergency SOS, and family location sharing — a substantial and clinically validated feature set [7], [51], [52]. The overlap between this feature set and SENTRIX CORE's sensor-level capabilities (heart rate, SpO₂, sleep, GPS, fall detection) is real and should not be understated.

The distinction is not primarily one of sensors, but of purpose. Apple Watch is designed, first, as a general-purpose smartwatch, with safety functions as one feature among many aimed at a broad consumer market. SENTRIX CORE is designed, first, as a personal-safety intelligence platform, for which the wearable device is one component among several (the mobile application of Section 12, the AI architecture of Section 6, and the multi-platform ecosystem of Section 12.7). This difference in purpose produces a difference in behavior that can be summarized concisely: Apple Watch's fall-detection and Emergency SOS workflow (Figure 7) answers the question “did something happen?” — it responds after an accelerometer- and gyroscope-based impact is detected, asks the user to confirm, and calls emergency services if there is no response [7]. SENTRIX CORE's Risk Score Engine (E9, Section 6) is designed to also ask a second question before any impact occurs: “is this person becoming unsafe?” — by continuously fusing motion, location behavior, heart rate, and audio context into the predictive and behavioral-deviation signals described in Engines E1b, E7, and E11.



Figure 7. Apple Watch's fall-detection and Emergency SOS interface. The workflow is reactive: a hard fall is detected, the user is asked to confirm they are okay, and Emergency SOS is offered as the escalation path if there is no response.



Figure 8. Apple Watch's rear sensor array (optical heart-rate and ECG electrodes). The physical sensing modality — optical PPG for heart rate — is comparable to the MAX86176 sensor described in Section 13.4; the difference SENTRIX CORE proposes lies in the AI decision layer built on top of this class of sensor, not in the sensor itself.

Apple Watch: Fall occurs → impact detected → ask “Are you okay?” → no response → Emergency SOS.

SENTRIX CORE: Continuous multi-modal monitoring → early deviation detected (motion, location, heart rate, audio, personal baseline) → Risk Score rises before any impact → Smart Confirmation (Section 7.2) → graduated escalation if risk is confirmed or unanswered.

The clearest single expression of this distinction is a positioning statement rather than a feature list: *Apple Watch detects emergencies after they happen; SENTRIX CORE is designed to predict and, where possible, help prevent emergencies before they happen.* Scenario 20 in Section 9 (pre-fall prediction) and the predictive fatigue warning in Scenario 9 illustrate this distinction concretely: in both cases, the relevant AI engine (E1b or E11) acts on a rising-risk trend several seconds to several minutes before the event that a purely reactive system would instead have to detect after the fact.

3.2 Commercial Products and Startups

This section compares SENTRIX CORE against the broader commercial wearable and safety landscape, organized into three market clusters: smart consumer wearables (Apple Watch, Garmin, Fitbit, WHOOP, Oura), family and location-safety applications (Life360, AngelSense), and medical/emergency alert platforms (Medical Guardian, Empatica, Biofourmis, Current Health).

Apple Watch (Apple Inc.)

Capabilities: Heart-rate monitoring, ECG, blood-oxygen monitoring, fall detection, Emergency SOS, crash detection, and location sharing [7], [51], [52].

Strengths: Massive consumer adoption, clinical validation, high-quality sensors, and mature emergency infrastructure.

Limitations: No continuous AI Digital Twin, no multi-modal threat intelligence, no predictive danger estimation, and no guardian ecosystem for multiple vulnerable users. See Section 3.1 for a detailed positioning comparison.

Fitbit (Google)

Capabilities: Activity tracking, heart-rate monitoring, sleep analysis, stress estimation, and ECG [8], [53].

Strengths: Large longitudinal user base; data has supported multiple cardiovascular and sleep-research studies.

Limitations: Focused on wellness rather than autonomous safety; no emergency AI reasoning, predictive danger estimation, or standalone communication.

Garmin (Garmin Ltd.)

Capabilities: GPS, heart-rate monitoring, fitness metrics, stress tracking, and training analysis [9], [54].

Strengths: Excellent outdoor tracking and sports analytics.

Limitations: Designed primarily for athletes rather than elderly protection, personal security, or AI emergency response.

WHOOP (WHOOP Inc.)

Capabilities: Continuous heart-rate monitoring, HRV, recovery score, sleep coaching, and strain analysis [55].

Strengths: Demonstrated the practical value of continuous physiological monitoring, particularly HRV-based recovery scoring.

Limitations: No GPS-based emergency features, SOS, guardian system, or threat detection; optimized for athletic performance rather than danger prevention.

Oura Ring (Oura Health)

Capabilities: Sleep monitoring, temperature trends, HRV, and recovery scoring [56].

Strengths: Strong passive health tracking in an unobtrusive ring form factor.

Limitations: No emergency response, real-time safety monitoring, standalone connectivity, or motion intelligence; sensor count is constrained by the ring form factor.

Life360 (Life360 Inc.)

Capabilities: Family location circles, driving-behavior safety scoring, crash detection (phone-based), and location-based alerts.

Strengths: Among the closest existing competitors in spirit: a large, established family-safety user base and strong location-sharing infrastructure.

Limitations: No dedicated wearable hardware, no physiological sensing, and no AI-driven behavioral or threat prediction; the guiding question is “where is my family member?” rather than “is my family member becoming unsafe?”

Medical Guardian (Medical Guardian)

Capabilities: SOS button, fall detection, GPS, and a professional emergency-response call center [57].

Strengths: Mature, professional emergency-response infrastructure targeted at elderly safety.

Limitations: A traditional Personal Emergency Response System (PERS): emergency occurs, the user presses a button, and help arrives. No AI prediction, Digital Twin, or behavioral learning.

AngelSense (AngelSense LLC)

Capabilities: GPS tracking, geofencing alerts, and caregiver communication for child safety [58].

Strengths: Purpose-built for child-monitoring use cases, with strong school-route tracking.

Limitations: No health AI, sensor fusion, or physiological intelligence, and no equivalent to the Digital Twin behavioral model described in Section 6 (Engine E7).

Empatica (EmbracePlus) (Empatica S.p.A.)

Capabilities: Electrodermal activity (EDA), skin temperature, accelerometry, sleep, and stress monitoring, used extensively in clinical and medical research [14], [59].

Strengths: Scientific-grade physiological sensing validated in clinical studies.

Limitations: A research and clinical wearable platform, not a consumer emergency ecosystem; no guardian network, GPS-based safety, or autonomous escalation logic.

Biofourmis (Biofourmis Inc.)

Capabilities: AI-driven healthcare analytics, remote patient monitoring, and predictive health scoring [60].

Strengths: Strong clinical AI capability for hospital-affiliated monitoring programs.

Limitations: Hospital-oriented deployment model, not a personal safety wearable for daily consumer use; no guardian network or nearby-helper mechanism.

Current Health (Current Health)

Capabilities: Remote monitoring, an AI healthcare platform, and clinician-facing patient dashboards [61].

Strengths: Integrated healthcare-provider ecosystem.

Limitations: Built for healthcare providers rather than personal, autonomous, everyday protection.

3.3 Academic Related Work Comparison

Research Area 1 — Fall Detection

Mubashir et al. [1] applied wearable sensors and machine-learning classification to automatic fall detection; the approach detects a fall only after it happens and does not incorporate prediction or personalization. Igual et al. [2] surveyed vision-based, wearable, and ambient fall-detection systems, concluding that no complete AI safety ecosystem existed at the time of their review.

Research Area 2 — Human Activity Recognition

Nweke et al. [10] reviewed CNN, RNN, and other deep architectures applied to human activity recognition from wearable and mobile sensors. This body of work is focused on activity classification rather than integrated safety intelligence.

Research Area 3 — Sensor Fusion

Sensor-fusion research combines inertial measurement units (IMUs), GPS, and physiological signals using Bayesian fusion, Kalman filtering, neural networks, and transformer-based fusion architectures [5], [6]. While fusion reliably improves robustness, most published systems target a single application domain rather than a general-purpose safety platform.

Research Area 4 — Digital Twin Healthcare

Fuller et al. [13] formalized a Digital Twin architecture consisting of a physical entity, data acquisition, a virtual model, and a prediction stage. Most implementations of this architecture remain industrial or clinical rather than targeted at personal wearable safety.

Research Area 5 — TinyML and Edge AI

Warden and Situnayake [11] describe methods for running machine-learning inference directly on microcontrollers and other embedded devices. For SENTRIX CORE, this line of work is directly relevant to achieving low latency, improved privacy, and offline intelligence on the wearable band itself, as implemented in the on-device AI pipeline described in Section 13.11.

3.4 Comparative Feature Matrix

Table 1 summarizes binary feature coverage across representative commercial systems and the proposed SENTRIX CORE architecture.

System	Health	Motion AI	GPS	SOS	Fall Detection	Digital Twin	Predictive AI	Multi-user Guardian	Edge AI
Apple Watch	✓	Limited	✓	✓	✓	✗	Limited	✗	Limited
Fitbit	✓	✓	Limited	✗	✗	✗	Limited	✗	✗
Garmin	✓	✓	✓	Limited	Limited	✗	Limited	✗	✗
WHOOP	✓	Limited	✗	✗	✗	Limited	✓	✗	✗
Oura	✓	✗	✗	✗	✗	Limited	Limited	✗	✗
Life360	✗	Limited	✓	Limited	✗	✗	Limited	✓	✗
Medical Guardian	✗	Limited	✓	✓	✓	✗	✗	Limited	✗
Empatica	✓	✓	✗	✗	✗	✗	✓	✗	✓
SENTRIX CORE	✓	✓	✓	✓	✓	✓	✓	✓	✓

Table 1. Binary feature comparison of representative wearable and safety systems against the proposed SENTRIX CORE architecture.

Table 2 complements this binary view with a graded 0–5 assessment across six broader capability dimensions, in order to represent differences of degree (e.g., partial versus comprehensive health monitoring) that a binary matrix cannot capture.

Product	Health Monitoring	Safety Features	AI Prediction	Guardian Ecosystem	Digital Twin	Threat Detection
Apple Watch	5	3	0	2	0	0
Garmin	4	2	0	0	0	0
Fitbit	4	2	1	0	0	0
WHOOP	5	0	1	0	0	0
Oura	5	0	1	0	0	0
Life360	1	4	2	5	0	1
AngelSense	1	4	1	3	0	1
Medical Guardian	3	4	0	3	0	0
SENTRIX CORE (proposed)	4	5	5	5	5	5

Table 2. Graded comparison (0 = absent, 5 = comprehensive) across six capability dimensions. SENTRIX CORE's health-monitoring score reflects a deliberate scope decision (Section 3.6): the architecture prioritizes safety-relevant physiological signals (Section 6, Engine E3) over the broader wellness-tracking feature set (e.g., detailed fitness coaching) offered by Apple Watch, WHOOP, and Oura.

3.5 Research Gap Summary

Reviewing academic papers, commercial products, and startups jointly indicates that no existing system combines: (1) a personal Digital Twin, (2) multi-modal sensor fusion, (3) predictive risk intelligence, (4) autonomous emergency response, (5) standalone wearable hardware, and (6) a guardian ecosystem, within a single product. Life360 comes closest on the guardian and location dimensions but has no wearable sensing layer; Apple Watch comes closest on sensing and reactive safety but has no predictive or guardian-ecosystem layer; Empatica comes closest on physiological sensing depth but is not a consumer safety product.

3.6 SENTRIX CORE Positioning

SENTRIX CORE is positioned between current smartwatches — which combine health tracking with notifications — and medical safety systems — which combine an emergency button with passive monitoring. It introduces a continuous pipeline: sensing, personal AI Digital Twin modeling, risk prediction, an autonomous decision engine (Section 7), and emergency response.

The individual technological building blocks reviewed above — wearable sensing, AI classification, fall detection, Digital Twins, and GPS-based safety — are each independently mature. The gap that remains is integration: a unified AI-powered autonomous personal-safety intelligence platform. Consistent with the market-positioning argument in Section 3.1, SENTRIX CORE is deliberately not positioned as a competing smartwatch; it is positioned as a category-level proposal — an AI personal-safety intelligence platform — for which a wearable device, a mobile application, and a set of institutional dashboards (Section 12.7) are the delivery mechanism rather than the product itself.

4. Research Gap Analysis

This section synthesizes the literature and market review into five concrete research gaps and explains how the proposed architecture addresses each one.

4.1 Existing Technology Landscape

Current solutions can be grouped into five categories.

Category 1 — Smart Consumer Wearables

Apple Watch, Fitbit, Garmin, and Samsung Galaxy Watch provide heart rate, sleep monitoring, activity recognition, ECG (on some devices), and fall detection. These systems answer “what is happening now?” but not “what is likely to happen next?” — they are detection systems rather than prediction systems.

Category 2 — Personal Emergency Response Systems (PERS)

Medical Guardian, Life Alert, and MobileHelp follow a linear architecture in which the user presses an SOS button, a call center is contacted, and emergency response follows. These systems depend on user awareness and manual activation, and cannot detect silent emergencies, forced situations, or gradual health deterioration.

Category 3 — Medical Remote Monitoring Platforms

Biofourmis, Current Health, and Philips RPM provide vital-sign monitoring, patient analytics, and clinical dashboards, but are designed for hospitals and healthcare providers rather than daily personal protection, child safety, elderly independence, or outdoor emergencies.

Category 4 — Research Wearable AI Systems

Academic research on fall detection, human activity recognition, stress detection, and emotion recognition commonly applies CNNs, LSTMs, transformers, and classical machine learning, but typically solves one isolated problem (e.g., a fall detector) rather than a complete safety intelligence system.

Category 5 — Digital Twin Systems

Digital Twin research provides personalized models, prediction, and simulation for manufacturing, healthcare, and smart cities. Healthcare Digital Twins in particular are usually hospital-based and cloud-dependent rather than real-time wearable systems.

4.2 Identified Research Gaps

Gap 1 — Lack of a Unified Safety Intelligence Architecture

Current systems follow a shallow pipeline: a single sensor feeds a single algorithm, which produces an alert (e.g., accelerometer → fall detection → notification). The architecture required for robust safety intelligence is deeper: multiple sensors feed multiple AI engines, whose outputs are fused, interpreted through a personal Digital Twin, converted into a risk prediction, evaluated by a decision engine, and — where warranted — escalated to an emergency response. SENTRIX CORE's contribution here is a unified, multi-layer AI safety architecture rather than an isolated algorithm.

Gap 2 — Detection Instead of Prediction

Most systems detect that a fall has already happened, that heart rate is already abnormal, or that the user has already pressed SOS. The proposed Predictive Risk Score Engine instead estimates the probability of an adverse event before it occurs, for example by combining a changing motion pattern, a decreasing HRV, rising stress indicators, and a location anomaly into a single risk score (e.g., 82% — high risk — prepare emergency response).

Gap 3 — Lack of Personalization

Current models are trained on population-level data (e.g., “average human walking behavior”), which does not account for individual variation. SENTRIX CORE addresses this with a Personal AI Digital Twin that learns a physical profile (walking style, activity level, resting heart rate), a behavioral profile (daily routine, location patterns, sleep behavior), and a health profile (HR baseline, stress response, recovery pattern), continuously updated as the user's own sensor history accumulates.

Gap 4 — Lack of Multi-Modal Intelligence

Most systems rely on a single modality — for example, fall detection from accelerometer data alone. Reliable human-safety inference instead requires combining motion (fall, impact, movement), health (HR, HRV, SpO₂), audio (voice, ambient environment), and location (GPS, geofenced context). SENTRIX CORE's Multi-Modal Sensor Fusion Engine combines motion, health, audio, behavior, and location AI outputs through a fusion network to produce a unified risk assessment.

Gap 5 — Lack of Autonomous Emergency Intelligence

In existing systems, a human must trigger the emergency response. SENTRIX CORE is designed so the AI decision layer can autonomously decide to escalate. For example, if the system detects an impact, an absence of subsequent movement, an abnormal heart rate, and no user response, it can compute an emergency-confidence score (e.g., 94%) and autonomously activate guardian alerting, GPS sharing, audio-evidence capture, and emergency communication.

5. SENTRIX CORE Scientific Contribution

The primary contribution of this work is not the wearable hardware itself — numerous companies already manufacture smart bands and watches. The contribution proposed here is an autonomous AI safety-intelligence layer composed of four complementary elements.

Contribution 1 — AI Safety Fusion Architecture

A hierarchical architecture spanning a sensor layer, an edge-processing layer, individual AI engines, a fusion layer, a risk engine, and an emergency-decision layer, allowing each stage to be developed, validated, and updated independently while still contributing to a single coherent decision.

Contribution 2 — Personalized Safety Digital Twin

Unlike static models that are trained once and deployed indefinitely, the SENTRIX CORE Digital Twin is deployed, observes the user, learns their behavior, updates its internal model, and thereby improves the quality of its decisions over time.

Contribution 3 — Context-Aware Risk Prediction

Rather than a fixed rule such as “if acceleration exceeds a threshold,” the system jointly considers motion, health, time of day, location, personal history, and behavior. For instance, a high heart rate combined with fast movement is normal for a person who is running, while the same high heart rate combined with abnormal movement during typical sleeping hours may indicate a possible emergency.

Contribution 4 — Edge-Cloud Hybrid Intelligence

Most AI wearable systems rely exclusively on cloud inference, introducing latency, privacy, and connectivity risk. SENTRIX CORE instead performs real-time inference — fall detection and emergency decisions — on the edge (the wearable band), while reserving the cloud for computationally heavier tasks such as model improvement, longitudinal analytics, and Digital Twin updates, coordinated through a companion mobile guardian application.

6. AI Engine Architecture (E1–E13)

The proposed SENTRIX CORE decision layer is composed of thirteen cooperating AI engines, each responsible for a distinct sensing modality or system function. Engines execute on the wearable band, a companion smartphone, and/or the cloud, depending on their latency and compute requirements. Outputs from the individual engines are combined by the Risk Score Engine (E9) into a single, unified decision.

E1 Fall Detection AI

Detects falls, hard impacts, collapses, slips, seizure-like motion, and post-fall inactivity. The most safety-critical of the reactive engines.

Inputs	BMI270 accelerometer X/Y/Z ($\pm 16g$, 400Hz); BMI270 gyroscope X/Y/Z (± 2000 dps, 400Hz); time-series IMU windows (2.5s sliding, 50% overlap).
Outputs	Fall probability (0.0–1.0); fall-type classification (fall / hard impact / collapse / slip / seizure / inactivity); confidence score; post-fall inactivity duration.
Training datasets	SisFall (19 fall types, 15 ADL types, elderly subjects) [63]; MobiAct (9 fall types) [64]; UMAFall (wrist-worn, pre-impact IMU) [65]; Opportunity (activity context for false-positive reduction).
Performance	Latency < 5 ms (band) / < 80 ms (phone). Reported accuracy > 95% on a held-out SisFall test split, with F1 > 0.94 on the fall class.

Model Architecture

Platform	Model	Notes
Band (edge)	Tiny 1D CNN	3 conv layers (8/16/32 filters), kernel size 5, GlobalAvgPool, 2-class softmax; < 32 KB RAM
Phone	CNN + LSTM	CNN feature extractor (32→64→128) → LSTM(128) → Dense → Softmax; TFLite INT8

Illustrative scenario. *An elderly user, home alone: a sudden acceleration spike is detected at 09:47, orientation changes from vertical to horizontal, and no motion follows for 12 seconds. Heart rate remains present at 64 BPM. The engine outputs a fall probability of 0.97; combined with home/night/alone context, this contributes 25 of 25 possible points to the unified risk score, and an emergency workflow is activated.*

E1b

Early Fall Prediction AI (Pre-Fall Detection)

Detects balance instability two to five seconds before a fall occurs — a proactive, rather than reactive, fall-prevention capability.

Inputs	BMI270 IMU at 400Hz in 2–5 second sliding windows; body-orientation change rate; gait symmetry metrics; center-of-gravity deviation estimate.
Outputs	Pre-fall risk score (0.0–1.0); instability type (postural / gait / acceleration / center-of-gravity); estimated time to impact; confidence score.
Training datasets	SisFall (pre-fall motion sequences) [63]; UMAFall (pre-impact IMU) [65]; MobiAct (balance instability leading to falls) [64]; a custom synthetic dataset from controlled force-plate experiments.
Performance	Latency < 5 ms (band) / < 60 ms (phone). Reported sensitivity of approximately 70% at t–2s and 88% at t–1s, with a false-positive rate below 8% once workout context is suppressed by the Context Engine (E8).

Model Architecture

Platform	Model	Notes
Band	Tiny 1D CNN + rules	Lightweight instability detection; < 5 ms
Phone (primary)	CNN-LSTM	Per-frame CNN extraction → LSTM(128) temporal degradation learning; < 60 ms
Phone (alt.)	TCN	Dilated causal convolutions for efficient long-context analysis; < 60 ms

Illustrative scenario. At home, IMU data shows irregular gait symmetry (34% stride-asymmetry deviation from baseline). Pre-fall risk reaches 0.45 at t–3s, then 0.73 at t–2s as lateral sway increases, triggering a silent vibration alert; by t–1s risk reaches 0.91 and the emergency stack is pre-activated. When the fall occurs at t–0, the standard Fall AI (E1) takes over confirmation.

E2 Human Activity Recognition AI

Continuously classifies the user's current activity state, forming the contextual foundation for all other AI interpretation and for Digital Twin behavior modeling.

Inputs	BMI270 IMU at 400Hz, downsampled to 50Hz; 5-second sliding windows, 50% overlap.
Outputs	Activity class (walking / running / standing / sitting / lying / sleeping / driving / workout — 8 classes); activity confidence; activity duration.
Training datasets	WISDM (smartphone + smartwatch, 36 subjects) [66]; UCI HAR (30 subjects, 6 activities) [67]; PAMAP2 (9 activities, 18 subjects, 52 Hz) [68]; RealWorld HAR (wrist-worn position variation).
Performance	Latency < 5 ms (band) / < 80 ms (phone). Reported accuracy > 95% on the WISDM test split, with walking/running/driving classification > 98% and workout-vs-fall disambiguation > 97%.

Model Architecture

Platform	Model	Notes
Band	Tiny CNN	8-class classifier; < 20 KB RAM; < 5 ms
Phone	CNN-LSTM (best)	CNN (32→64→128) + LSTM(256) + Attention + Dense; TFLite INT8; < 80 ms

Illustrative scenario. An athlete's accelerometer shows high-G, rapid direction-change motion. The Activity AI classifies this as "Running — Workout" at 94% confidence for a 23-minute duration. This context is passed to the Context Engine, which reduces fall-detection sensitivity, so a subsequent heart-rate spike to 178 BPM is correctly interpreted as normal for high-intensity running rather than an emergency.

E3

Health Anomaly AI

Continuously monitors cardiac and physiological signals, detecting heart-rate anomalies, HRV collapse, SpO₂ drops, stress indicators, and elevated cardiac-event risk.

Inputs	MAX30102 raw PPG waveform (50–200Hz); derived heart rate (BPM); HRV (RMSSD, SDNN, LF/HF ratio); SpO ₂ (%).
Outputs	Health risk score (0.0–1.0); anomaly type (hr_high / hr_low / hrv_collapse / spo2_drop / stress / fatigue / cardiac_risk); deviation from personal baseline (Z-score); confidence score.
Training datasets	MIMIC-III (ICU clinical data, arrhythmia and cardiac events) [69], [80]; PhysioNet (ECG/PPG signal library) [69]; PPG-DaLiA (wearable PPG in daily-life activities) [70]; WESAD (stress and affect detection) [71].
Performance	Latency < 1 ms (band rule engine) / < 50 ms (phone) / < 500 ms (cloud). The personalized autoencoder achieves reported sensitivity of 94% for cardiac anomaly detection, with a false-positive rate below 3% relative to the user's own baseline.

Model Architecture

Platform	Model	Notes
Band	Rule-logic engine	HR <40 or >180 BPM → flag; SpO ₂ <90% → critical; HRV delta >40% → alert; < 1 ms
Phone	Personalized LSTM autoencoder	Trained on the user's own 7-day baseline; reconstruction error = anomaly score; TFLite INT8; < 50 ms
Cloud	XGBoost population model	Cross-population patterns and comorbidity risk; periodic retraining; < 500 ms

Illustrative scenario. A 65-year-old user with known hypertension has a sleep-time baseline heart rate of 58 BPM. Over four minutes at 03:22, heart rate rises from 58 to 148 BPM, HRV RMSSD collapses from 45 ms to 12 ms, and SpO₂ falls from 97% to 91%. The autoencoder's reconstruction error reaches 3.8× its threshold, producing a health risk score of 0.93 and triggering a medical-emergency data packet containing the user's cardiac profile.

E4

Distress Audio AI

Detects acoustic signatures associated with emergencies: screams, cries, help-seeking vocalizations, sounds of violence, and general panic audio patterns.

Inputs	INMP441 event-triggered audio bursts (3–10 seconds); MFCC (40 coefficients); mel-spectrogram (96×64 log-mel patches); 1-second frame analysis.
Outputs	Distress score (0.0–1.0); audio class (scream / cry / help_vocalization / violence_sound / panic / normal); event confidence; duration of distress event.
Training datasets	AudioSet (632 audio-event classes, 2M clips, Google) [72]; ESC-50 (2,000 environmental clips, 50 classes) [73]; UrbanSound8K (8,732 urban-environment clips) [74]; CREMA-D (emotion-relevant audio, 91 actors) [77].
Performance	Latency < 15 ms on-device (YAMNet) / < 200 ms cloud (wav2vec2). Reported precision of 89% (YAMNet) and 94% precision / 91% recall for the fine-tuned wav2vec2 model.

Model Architecture

Platform	Model	Notes
Phone	YAMNet (Google)	521-class audio classification, 1s frames; TFLite FP16; < 15 ms on mobile NPU
Cloud	wav2vec2	Self-supervised audio transformer, fine-tuned on a custom distress dataset; < 200 ms

Illustrative scenario. A woman alone in a parking lot at 22:30 triggers a motion anomaly, which in turn triggers an audio capture. YamNet detects a scream (0.88) and a female distress vocalization (0.76), producing a distress score of 0.91 and contributing 18.2 of 20 possible points to the unified risk score. Emergency escalation is triggered, and a 5-second audio clip is SHA-256 hashed and stored as evidence.

E5

Safe Word + Panic Voice AI

Analyzes emotional state from voice tone, allowing the system to override a user's verbal claim of being safe when panic biomarkers are present in the voice signal.

Inputs	Phone microphone, 3-second sample captured on a confirmation prompt; pitch (F0), jitter, shimmer, harmonics-to-noise ratio (HNR), spectral centroid, breathiness; the user's personal 7-day voice baseline.
Outputs	Emotional state (calm / stress / fear / panic / neutral); panic probability (0.0–1.0); voice-override flag; deviation from personal voice baseline.
Training datasets	IEMOCAP (12 hours, 10 actors, fine-grained emotion labels) [75]; RAVDESS (24 actors, 7,356 files, 8 emotional expressions) [76]; CREMA-D (7,442 clips, 91 actors) [77].
Performance	Latency < 200 ms on mobile NPU. Reported emotion-classification accuracy of 78% on the IEMOCAP test set, and 91% accuracy for the binary fear-versus-calm distinction.

Model Architecture

Platform	Model	Notes
Phone	Wav2Vec2 (emotion)	Self-supervised transformer fine-tuned for emotional state; TFLite INT8; < 200 ms
Phone (alt.)	HuBERT	Alternative self-supervised model with comparable accuracy

Illustrative scenario. A user verbally responds “I’m okay” to a confirmation prompt. Semantic analysis of the words alone is positive, but acoustic analysis shows fear at 0.87, panic at 0.79, jitter deviating 2.3× from the personal baseline, and HNR degraded by 40%. The verbal claim is overridden, Silent Emergency Mode is activated, and the system continues escalation while outwardly appearing to accept the user’s response.

E6Attack / Kidnapping Detection AI

Detects coercion, physical attack, kidnapping, and violent threat through simultaneous six-stream multimodal analysis.

Inputs	Motion (BMI270 violent-motion signatures); audio (INMP441 via YAMNet distress scores); GPS (route deviation and forced movement); health (MAX30102 panic HR + HRV); behavior (Digital Twin deviation score); phone state (BLE disconnection, silence, SOS suppression).
Outputs	Threat probability (0.0–1.0); threat type (physical_attack / kidnapping / coercion / forced_movement); confidence score; recommended mode (silent_emergency / critical_emergency).
Training datasets	A fusion of all sensor-specific datasets listed above; a custom synthetic dataset of attack/kidnapping motion profiles generated through physical simulation.
Performance	Latency < 100 ms (phone) / < 500 ms (cloud). Reported precision of 91% and recall of 87% for violent-attack scenarios on a synthetic test set, with a false-positive rate below 1% in daily use.

Model Architecture

Platform	Model	Notes
Phone	Fusion network	Parallel per-modality encoders → concatenation → dense fusion → threat probability; < 100 ms
Cloud	Multimodal transformer	Cross-attention across all modalities for retrospective, highest-accuracy analysis; < 500 ms

Illustrative scenario. A sudden violent motion spike (an arm jerk and struggle) coincides with a heart rate of 165 BPM, a BLE disconnection consistent with the phone being taken, a YAMNet distress score of 0.88, and a Digital Twin route-deviation flag active for 15 minutes. The fused threat probability reaches 0.94, triggering Silent Emergency Mode: no audible alert, covert GPS streaming every 5 seconds, a silent guardian alert, and an automatically generated police data packet.

E7 Digital Twin AI

Learns each user's unique behavioral patterns and detects meaningful deviations — the personalized behavioral-intelligence layer underlying the rest of the system.

Inputs	24/7 GPS traces, clustered via HDBSCAN; motion patterns from Activity AI (E2) over time; health baselines (daily HR, HRV, SpO ₂ norms); time patterns (wake / sleep / activity hours); sleep architecture inferred from motion and heart rate.
Outputs	Behavioral deviation score (0.0–1.0); deviation type (unusual_location / unusual_time / unusual_activity / unusual_route); deviation magnitude; expected-versus-actual behavior description.
Training datasets	GeoLife GPS (Microsoft Research, 182 users, 5 years) [78]; the user's own personal health data (self-supervised); a 72-hour initial calibration period.
Performance	Latency < 100 ms per inference; HDBSCAN clusters are refreshed in batch every 6 hours. Reported wandering-detection sensitivity of 92%, with a 4% false-positive rate in deployment scenarios.

Model Architecture

Platform	Model	Notes
Phone	LSTM autoencoder	Behavioral embedding; reconstruction error = deviation score; TFLite INT8; < 100 ms
Phone	HDBSCAN clustering	Learns frequent locations (home / work / school) without manual labeling
Phone	LSTM trajectory model	Predicts the expected route to detect deviation

Illustrative scenario. For a 72-year-old Alzheimer's patient, the Digital Twin expects sleep at 02:47 AM (confirmed in 73 of the previous 74 readings at that time). Instead, the sensors show walking, GPS moving away from home at 4.2 km/h. The behavioral deviation score reaches 0.91, combining unusual location, time, and activity; the combined risk score reaches 78 (HIGH_RISK), and a guardian is alerted that the patient is wandering roughly 400 m from home.

E8 Context / Decision Engine

Interprets the current situation and adjusts the interpretation of every other AI engine's output, substantially reducing false alarms.

Inputs	Time of day; current location type (safe_zone / unknown / dangerous / indoor / outdoor); Activity AI classification; user account type and profile; device state (battery, BLE connectivity); recent event history (last 60 minutes).
Outputs	Context adjustment factor (0.0–2.0, a multiplier applied to risk scores); a plain-text situation description; a recommended sensitivity adjustment.
Training datasets	Derived from the context associated with all other engine datasets; a custom set of annotated context-outcome pairs drawn from real emergency events.
Performance	Latency < 1 ms. Context-adjusted models reduce the false-alarm rate by a reported 62% relative to a context-naïve baseline.

Model Architecture

Platform	Model	Notes
Phone	LightGBM	Fast gradient boosting with explainable feature importance; < 1 ms
Phone (alt.)	Rules + XGBoost	A hybrid approach for edge cases where simple rules fail

Illustrative scenario. An athlete at 17:30, in a recognized gym location, classified by the Activity AI as “Workout,” triggers a high-G motion event that the Fall AI scores at 0.72 fall probability. Without the Context Engine, this would raise the unified risk score to 72 (HIGH_RISK). With the Context Engine applying a context factor of 0.35 (gym + workout + known personal pattern), the adjusted score falls to 25 (SAFE), avoiding a false alarm; the context factor returns to 1.0 once the workout ends.

E9

Risk Score Engine

Aggregates every AI engine's sub-score into a single unified Risk Score (0–100) that drives all emergency decisions.

Inputs and weights	Fall Score (E1): 25%; Health Score (E3): 20%; Audio Score (E4): 20%; GPS Risk Score (E10): 15%; Behavioral Deviation (E7): 10%; Context Factor (E8): 10%.
Outputs	Unified risk score (0–100); risk status (SAFE / WARNING / HIGH_RISK / CRITICAL); dominant contributing factor; overall system confidence.
Training datasets	Labeled emergency events drawn from all constituent sensor datasets; calibration against confirmed emergency outcomes versus predicted scores.
Performance	Latency < 5 ms. Reported true-emergency detection rate above 97%, with a production false-alarm rate below 2%.

Model Architecture

Platform	Model	Notes
Primary	XGBoost (weighted fusion)	Gradient boosting on weighted sensor inputs; < 5 ms
Research	Bayesian fusion	$\log P(\text{Emergency} \mid \text{all sensors})$ — a rigorous probabilistic alternative
Research	Resilient confidence fusion	Auto-weighted by Sensor Reliability AI (E12) scores; self-healing under sensor degradation

Illustrative scenario. At night, at home: Fall ($0.96 \times 25 = 24$) + Health ($0.90 \times 20 = 18$) + Audio ($0.00 \times 20 = 0$) + GPS ($0.80 \times 15 = 12$) + Behavior ($0.80 \times 10 = 8$) + Context ($0.70 \times 10 = 7$) yields a combined risk score of 69 (HIGH_RISK). A Smart Confirmation prompt is triggered with a 10-second timeout; if the user does not respond, the system escalates automatically to CRITICAL.

E10GPS + Safe Zone AI

Provides location intelligence: geofencing, route-deviation detection, unsafe-zone detection, wandering alerts, and missing-person indicators.

Inputs	GPS/GNSS coordinates (NMEA sentences); Kalman-filtered position estimates; user-defined safe zones; Digital Twin location clusters.
Outputs	GPS risk score (0.0–1.0); zone status (inside_safe / outside_safe / unsafe_zone); route-deviation score (0.0–1.0); trajectory prediction (expected next position).
Training datasets	GeoLife GPS (Microsoft Research, 182 users, 5 years) [78]; CRAWDAD mobility traces [79].

Model Architecture

Platform	Model	Notes
Phone	HDBSCAN clustering	Learns frequently visited locations without manual labeling
Phone	LSTM trajectory	Predicts the expected route from learned patterns
Phone	Kalman filter	Real-time GPS noise filtering and position estimation

Illustrative scenario. A user's real-time position is continuously compared against their learned safe zones and predicted route; a sustained deviation from the expected trajectory, combined with an unsafe-zone flag, contributes directly to the unified risk score computed by E9.

E11

Predictive Incident AI

A forward-looking engine that predicts danger before it occurs, combining rising heart rate with panic motion, an unsafe route, and audio distress to estimate the probability of a future incident.

Inputs	Combined outputs of E1–E10 over a rolling temporal window (HR trend, motion pattern, route context, audio distress).
Outputs	Incident probability score (e.g., 78%).
Training datasets	A hybrid internal dataset combining the outputs and outcomes of the other twelve engines.
Status	Forward-looking capability; MVP implementation uses a Temporal Convolutional Network (TCN), with a Temporal Transformer identified as a future, higher-capacity version.

Model Architecture

Platform	Model	Notes
MVP	TCN	Efficient temporal convolution over the fused engine outputs
Advanced (future)	Temporal Transformer	Higher-capacity sequence model for longer-range risk forecasting

Illustrative scenario. Rising heart rate, an emerging panic-consistent motion pattern, an unsafe route deviation, and early audio distress are jointly observed; the engine estimates a 78% probability of an impending incident before any single-engine threshold would independently trigger an alert, allowing the system to pre-position its response.

E12

Sensor Reliability AI

A production-critical engine that detects a detached band, a noisy signal, sensor drift, or sensor failure, and feeds a reliability score into the Risk Score Engine (E9).

Inputs	Raw signal quality metrics from each sensor stream (IMU, PPG, microphone, GPS).
Outputs	Sensor-validity flags, e.g., “PPG Invalid,” “Band Detached,” or sensor-drift warnings.
Training datasets	Synthetic fault-injection data initially, followed by real device logs collected in production.
Model architecture	Band: a lightweight rule engine; Phone: an Isolation Forest anomaly detector.

Illustrative scenario. If the PPG sensor reports a physically implausible signal consistent with the band being loose or detached, E12 flags the health stream as unreliable; the Resilient Confidence Fusion variant of E9 then automatically down-weights the Health Score (E3) contribution rather than generating a false alarm from corrupted data.

E13

AI Governance / Model Health Engine

An industrial-grade monitoring layer that detects model drift, accuracy degradation, and the need for a rollback of a deployed model version.

Inputs	Streaming statistics of model inputs and outputs across the fleet of deployed devices.
Outputs	Drift alerts (“Model Drift Detected”) and rollback recommendations.
Drift types monitored	Data drift (change in input distribution); concept drift (change in the input–output relationship); performance drift (degradation in accuracy or other metrics over time).
Model architecture	Statistical drift-detection methods applied continuously to production telemetry.

Illustrative scenario. If a firmware or sensor-hardware revision subtly changes the distribution of incoming IMU data, E13 detects the resulting data drift, flags the affected model version, and recommends a rollback or retraining cycle before the drift measurably degrades fall-detection accuracy in the field.

6.1 Complete Dataset Inventory

Table 2 consolidates every public benchmark dataset referenced across the thirteen AI engines, together with the domain and engine(s) each dataset supports.

Dataset	Domain	Engine(s)	Key Properties
SisFall	Fall Detection	E1, E1b	19 fall types, 15 ADL types, elderly subjects
MobiAct	Fall Detection	E1, E1b	9 fall types, wearable sensor data
UMAFall	Fall Detection	E1, E1b	Wrist-worn fall detection, pre-impact IMU
Opportunity	Activity Context	E1	Activity context for false-positive reduction
WISDM	Activity Recognition	E2	Smartphone + smartwatch, 36 subjects
UCI HAR	Activity Recognition	E2	30 subjects, 6 activities, benchmark dataset
PAMAP2	Activity Recognition	E2	9 activities, 18 subjects, 52 Hz
RealWorld HAR	Activity Recognition	E2	Wrist-worn position variation
MIMIC-III	Clinical Health	E3	ICU clinical data, arrhythmia, cardiac events
PhysioNet	Clinical Signals	E3	ECG/PPG signal library, clinical validation
PPG-DaLiA	Wearable PPG	E3	Wearable PPG in daily-life activities
WESAD	Stress / Affect	E3	Stress and affect detection with physiological signals
AudioSet	Audio Events	E4	632 audio classes, 2M clips (Google)
ESC-50	Environmental Audio	E4	2,000 clips, 50 classes
UrbanSound8K	Urban Audio	E4	8,732 clips, urban environment
CREMA-D	Emotion Audio	E4, E5	7,442 clips, 91 actors, multi-dimensional ratings
IEMOCAP	Speech Emotion	E5	12 hours, 10 actors, fine-grained emotion labels
RAVDESS	Speech Emotion	E5	24 actors, 7,356 files, 8 emotional expressions
GeoLife GPS	Location / Tracking	E7, E10	182 users, 5 years (Microsoft Research)
CRAWDAD	Mobility Traces	E10	GPS mobility traces

Table 2. Complete dataset inventory referenced by the SENTRIX CORE AI engines.

6.2 Hardware Sensor Specifications

Sensor	Type	Specifications	Used By
BMI270	IMU (Accel + Gyro)	$\pm 16g$ / ± 2000 dps, 400 Hz	E1, E1b, E2, E6
MAX30102	PPG (HR + SpO ₂)	50–200 Hz sampling	E3, E6
INMP441	MEMS microphone	Event-triggered, 3–10 s bursts	E4, E5, E6
GPS/GNSS	Location	NMEA sentences, Kalman-filtered	E6, E7, E10

Table 3. Hardware sensor specifications underlying the SENTRIX CORE engines.

7. Emergency Intelligence and Escalation Architecture

Sections 5 and 6 introduced the Risk Score Engine (E9) and Emergency Orchestrator at the level of individual AI engines. This section specifies the complete decision pipeline that connects a raw sensor anomaly to a fully resolved emergency response. The guiding design principle is that SENTRIX CORE never treats a detected anomaly as an emergency by default. Instead, every event passes through a graduated sequence of assessment, confirmation, and escalation stages, consistent with safety-critical design practice in aviation and automotive systems: the system detects, assesses context, attempts to confirm with the user, and only then decides whether and how to act. This minimizes false alarms while preserving the ability to escalate immediately when confidence is high or the user cannot respond.

Sensors → AI Detection Engines → Sensor Fusion → Risk Score Engine → Context Engine → Smart Confirmation → Emergency Decision Engine → Communication Manager → Guardian / Nearby Rescue Network / Emergency Services → Recovery and Learning

7.1 Emergency Severity Levels

Every event processed by the Risk Score Engine (E9) is assigned a continuous risk score from 0 to 100, which is then mapped onto six discrete severity levels. Each level has a distinct system behavior, ranging from silent background logging at Level 0 to immediate, non-negotiable escalation at Level 5.

Level	Name	Risk Score	System Behavior
0	Normal	0–20	Background monitoring only. Sensor data is used to update the Digital Twin (E7); no alert of any kind is generated.
1	Awareness / Low Risk	20–40	A minor deviation from the user's learned baseline is logged (e.g., an unfamiliar route, a slightly elevated heart rate). No alert is shown to the user; AI monitoring frequency increases modestly.
2	Warning	40–65	The system requests Smart Confirmation from the user (Section 7.2): a vibration alert and an “Are you okay?” prompt, with a bounded response window. Sensor sampling and GPS update frequency increase.
3	High Risk	65–85	The system prepares — without yet activating — the emergency stack: GPS is set to high-frequency streaming, the LTE modem moves to standby, and an emergency data packet is drafted. A Guardian Warning notification is sent. Confirmation is still requested, on a shortened timeout.
4	Critical Emergency	85–95	Emergency Mode is activated: continuous GPS streaming, LTE prioritized, Guardian alerting, evidence capture, and — where enabled — activation of the Nearby Rescue Network (Section 7.4). The system no longer waits for confirmation once this level is reached autonomously (as opposed to being reached through user-initiated SOS).
5	Life-Threatening / Silent Emergency	95–100	The highest severity level. Used for confirmed cardiac collapse, confirmed attack or kidnapping signatures, and unconscious-user scenarios. No confirmation is requested. Where an attacker may be present, the system enters Silent Emergency Mode (Section 7.3) rather than issuing an audible or visible alert.

Table 32. The six Emergency Severity Levels and the corresponding system behavior at each level.

Two properties of this model are important. First, escalation is ordinarily monotonic and gradual: an event typically moves from Level 0 upward only as corroborating evidence accumulates across engines, and can de-escalate back to Level 0 at any point if the user confirms safety. Second, several engines are permitted to override the graduated path entirely. The Attack/Kidnapping AI (E6), the Panic Voice override (E5), and a manual 3-second logo long-press all bypass the intermediate levels and move directly to Level 4 or 5, because in these specific cases the cost of a brief delay is judged to outweigh the cost of an occasional false positive.

7.2 Smart Confirmation System

Smart Confirmation is the mechanism that prevents the majority of anomaly detections from ever becoming false alarms. Rather than escalating immediately, the system asks the user to confirm their status, and interprets the response — or its absence — in context.

Confirmation Channels

On the phone, a notification is shown with three options: “I’m Okay,” “Need Help,” and “Emergency.” On the band, the capacitive touch logo doubles as a confirmation control governed by press pattern:

Gesture	Meaning
Single press	I am okay — cancel escalation and update the Digital Twin with the new pattern
Double tap	Uncertain / possibly need help — enter heightened Monitoring Mode without triggering SOS
Three rapid taps	Manual SOS — immediate Emergency Mode
Long press (3 seconds)	Silent Emergency Mode — covert escalation with no visible or audible signal
Long press (6 seconds)	Critical Emergency — highest-priority escalation with full Guardian and evidence activation
No response within timeout	Escalate to the next severity level; treated as a possible incapacitation signal

Table 33. Touch-logo gesture vocabulary used for Smart Confirmation on the band.

The interpretation of a given gesture is context-dependent rather than fixed: a single press during a Warning-level prompt means “I’m okay,” while the same single press during an active Critical Emergency is interpreted as a request to cancel that emergency, and can itself be overridden by the Panic Voice AI (E5) if the accompanying voice sample shows fear biomarkers inconsistent with the words spoken (see Section 6, Engine E5, and the coercion example in Section 9).

Adaptive Timeout

The confirmation window is not fixed; it adapts to account type and the nature of the detected event, since the acceptable delay before escalation differs sharply between, for example, a cardiac anomaly and a child’s route deviation.

Scenario	Typical Confirmation Timeout
Suspected fall (adult account)	10–20 seconds
Suspected fall (elderly account)	5 seconds
Heart-rate / SpO ₂ collapse (medical account)	3–5 seconds, with override of user “I’m okay” responses above a health-risk threshold
Child safe-zone exit	30 seconds (child accounts do not wait for HIGH_RISK before alerting the guardian)
Alzheimer / wandering protocol	20 seconds, with immediate CRITICAL escalation on unknown-location detection
Mild behavioral deviation	20–30 seconds

Table 34. Representative adaptive confirmation timeouts by account type and event category.

The system also adapts over time to each individual: if a user typically responds within three seconds and on a given occasion does not respond for twenty, the Risk Score Engine treats that widened gap itself as evidence of elevated risk, independent of the underlying sensor data.

7.3 No-Response Handling and Silent Emergency Mode

A lack of response is one of the most informative — and most ambiguous — signals the system receives. SENTRIX CORE does not treat silence as automatically dangerous; instead, it re-evaluates risk using context before escalating further.

Context During Non-Response	Interpretation
During sleep hours, no other anomaly	Low added risk; treated as expected non-response
Immediately following a confirmed fall	High added risk; supports escalation to Critical
Immediately following violent motion or a distress-audio detection	Very high added risk; supports immediate escalation to Silent Emergency
During recognized workout activity	Low added risk; Context Engine (E8) suppresses escalation

Table 35. Context-dependent interpretation of a missed confirmation response.

When the situation indicates a possible threat from another person — rather than a medical or accidental event — the system enters Silent Emergency Mode. In this mode, no vibration, sound, or visible screen change occurs, so that a potential attacker has no indication that an alert has been raised. Internally, the system nonetheless performs the full emergency sequence: the LTE modem is activated, GPS position is streamed at short, fixed intervals, a Guardian receives a silent push notification explicitly instructing them not to call the user's phone (since a ringing phone could reveal the alert to an attacker), a short evidence-audio buffer is captured, and a police-escalation-ready packet is assembled and held pending Guardian authorization.

7.4 Nearby Rescue Network

For time-critical events, the nearest source of help is frequently not an ambulance or a family member but another verified SENTRIX CORE user in physical proximity. The Nearby Rescue Network is an opt-in layer that can be activated at Level 4 (Critical Emergency) in parallel with Guardian and emergency-service notification, rather than only after those channels have been exhausted.

Helper Verification and Trust Scoring

Participation as a Nearby Helper is optional and requires identity verification (phone number, email, and optionally government ID). Each helper accumulates a Trust Score (0–100) that rises with confirmed, completed responses and falls with ignored requests, repeated cancellations, or misuse.

Helper Selection

When the network is activated, candidate helpers are ranked by a weighted Rescue Priority Score combining estimated time-to-arrival (40%), physical distance (30%), Trust Score (20%), and current availability (10%), so that the fastest reliable responder is contacted rather than simply the geographically nearest one.

$$\text{Rescue Priority Score} = 0.40 \times \text{ETA-based factor} + 0.30 \times \text{distance factor} + 0.20 \times \text{Trust Score} + 0.10 \times \text{availability factor}$$

Privacy-Preserving Disclosure

Before a candidate helper accepts a request, they see only the type of emergency, the approximate distance and direction, and an estimated time to arrival — never the affected user's name, photograph, or medical information. Precise location is disclosed only after the helper explicitly accepts, at which point Emergency Navigation Mode opens and both the affected user's Guardian and the dispatched helper receive live position updates until the helper reports arrival. If no helper accepts within a bounded search window (typically 30 seconds), the system proceeds directly to emergency-services escalation without further delay.

7.5 Emergency Decision Matrix

Table 36 consolidates the mapping from detected situation to system action across the severity levels defined in Section 7.1.

Situation	Level	System Response
Minor route deviation	1	Log only; update Digital Twin
Safe-zone breach (child or elderly)	2–3	Smart Confirmation; Guardian warning
Suspected fall, user responsive	2–3	Confirmation prompt; cancel on response
Suspected fall, no response	4	Emergency Mode; Guardian alert; Nearby Rescue Network
Confirmed cardiac anomaly	4–5	Medical Emergency Mode; bypasses standard confirmation
Confirmed attack / kidnapping signature	5	Silent Emergency Mode; police-escalation-ready packet
Alzheimer wandering, unknown location	3–4	Recovery Mode; live tracking; Guardian alert
Critical battery during an active escalation	—	Battery Override Protocol suspends all power-saving limits

Table 36. Emergency Decision Matrix mapping representative situations to severity level and system response.

7.6 Recovery and Post-Incident Learning

An emergency is not considered resolved merely because an alert was sent. The system remains in an active monitoring state — with elevated sensor sampling and GPS update frequency — until one of three resolution conditions is met: the user explicitly confirms safety, a Guardian or Nearby Helper confirms arrival and resolution, or an authorized responder marks the incident closed. On resolution, the system compiles an Incident Report (surfaced to the user as the Incident Summary screen described in Section 12.4) recording what was detected, which engines contributed to the decision, the full timeline, and the final outcome, and feeds this record back into the user's Digital Twin (E7) so that the specific pattern that triggered the event is correctly classified in future.

7.7 Forced Band Removal During an Active Emergency

Section 14.6 specifies the two-layer mechanism by which the band distinguishes an ordinary removal from a forceful one: a rule-based Wear Detection Module that determines only whether the band is currently on the wrist, and a Forceful Removal Detection AI, implemented as a submodule of the Motion AI Engine, that classifies a removal event as Normal Removal, Accidental Slip, Forceful Removal, or Tampering Attempt. This subsection specifies how that classification feeds back into the severity-level model of Section 7.1 — specifically, how the system's response differs depending on whether a forceful removal occurs during ordinary, low-risk use or during an already-elevated risk state.

Forceful Removal Under Ordinary Conditions

If the Forceful Removal Detection AI reports a forceful removal while the current Risk Score (Section 7.1) is at Level 0 or Level 1, the event is handled exactly as specified in Section 14.6: it is logged, the Guardian is notified through the standard tamper-alert channel, and the Risk Score is nudged upward modestly. No further escalation follows automatically, since a forceful-looking removal in isolation — for example, a band caught and yanked free while removing a jacket — is not, by itself, strong evidence of a threat to the user.

Forceful Removal During an Elevated or Active Emergency

The response is materially different if a forceful removal is detected while the Risk Score is already at Level 2 (Warning) or above, or while an emergency is already active. This combination — elevated risk followed immediately by the band being torn from the wrist — is treated as strong corroborating evidence of a hostile act, most notably an attacker attempting to disable tracking during an assault or abduction (the scenario that originally motivated this capability, and consistent with Case 2 in Section 9.1). Critically, the system's behavior is the opposite of what it would do for an ordinary removal: rather than standing down monitoring, it escalates.

Forceful Removal Detection AI fires → current Risk Score $\geq$ Level 2, or an emergency is already active → treat as an aggravating signal, not a stand-down signal → force the Risk Score to Level 4 or Level 5 regardless of prior confirmation status → execute the forced-removal response sequence below

The forced-removal response sequence executes immediately, without waiting for the Smart Confirmation window described in Section 7.2, since the device that would carry the confirmation prompt is the one being removed:

- The last known GPS coordinates from both the band and, if still in range, the paired phone are locked in and transmitted before the connection is lost, rather than waiting for the next scheduled position update.
- The Risk Score is forced to Critical Emergency or Silent Emergency (Section 7.1) regardless of what confirmation state the event was in beforehand.
- The Guardian receives a distinct notification template that explicitly states the band was forcibly removed — rather than the generic fall or health-emergency wording — so that the nature of the event is unambiguous.
- If the paired phone remains in range and the account is not in standalone-only mode, protection continues through Phone-Only Mode (Section 12.3), using the phone's own motion, audio, and GPS sensing to maintain situational awareness after the band itself has gone offline.

- The evidence buffer — the seconds of motion, audio, and heart-rate data immediately preceding the disconnection — is finalized and transmitted immediately rather than held for the standard end-of-incident package (Section 7.6), since the moment of removal is itself evidentiary.

This design keeps the forced-removal mechanism consistent with the context-aware interpretation principle already established for unanswered confirmation prompts in Section 7.3: the same underlying sensor event (the band leaving the wrist) is deliberately interpreted differently depending on the risk context in which it occurs, rather than triggering one fixed response regardless of circumstance.

8. Nearby Rescue Network — Extended Specification

Section 7.4 introduced the Nearby Rescue Network as part of the Emergency Intelligence architecture. This section provides the full specification: enrollment and verification, the Trust Score and helper-selection model, staged privacy disclosure, in-field coordination, permission tiers, and multi-helper task assignment. The design rationale is straightforward: professional emergency services are highly capable but rarely the fastest responder in absolute time — a verified person located 100–300 meters away can frequently reach an incident in one to five minutes, well inside the typical ambulance response window in dense urban areas, and can in many cases provide meaningful assistance (presence, reassurance, basic first aid, or simply confirming the victim's status) before professional responders arrive.

8.1 System Rationale and Position in the Escalation Model

The Nearby Rescue Network is not a replacement for Guardian notification or emergency-service dispatch; it runs in parallel with both. Consistent with Section 7.1, it is activated starting at Severity Level 4 (Critical Emergency) rather than only at Level 5, because at Level 4 the system already holds high confidence that the user needs immediate help, and delaying community-network activation until Level 5 would discard exactly the time window in which a nearby helper is most likely to reach the user before professional services do. At Level 5 (Life-Threatening), the network activates simultaneously with emergency-service dispatch rather than sequentially before or after it.

AI detection → Risk Engine (E9) → Risk Score ≥ 85 → Critical Emergency Mode → user verification attempted → no qualifying response → Guardian alert issued → Nearby Rescue Network activated in parallel → emergency services engaged if unresolved

8.2 Opt-In Enrollment and Helper Verification

Participation is opt-in: any SENTRIX CORE user may choose to become an Emergency Helper, but no user is enrolled by default. Before a helper account is activated, the platform performs identity verification comprising phone-number verification, email verification, and — optionally, to unlock the Medical Helper tier described in Section 8.6 — government-ID verification. This verification step exists specifically to prevent the network from being used as a vector for locating vulnerable users under the guise of offering help.

8.3 Trust Score Model

Every enrolled helper carries a Trust Score on a 0–100 scale, which both gates eligibility for dispatch and weights helper selection (Section 8.4). The score moves upward on confirmed, completed responses and moves downward on ignored requests, repeated cancellations, or evidence of misuse.

Factor	Effect on Trust Score
Responded to a previous emergency request	Positive
Confirmed arrival at the incident location	Positive
No history of system misuse	Positive
Ignored a dispatch request	Negative
Repeated cancellations after accepting	Negative
Confirmed false or malicious report	Negative

Table 43. Positive and negative factors contributing to the Nearby Helper Trust Score.

8.4 Rescue Priority Scoring and Helper Selection

When the network is activated, every candidate helper within the search radius is ranked by a weighted Rescue Priority Score rather than by raw distance alone, because the geographically nearest helper is not always the fastest to arrive (a helper 100 m away on the far side of a blocked road may be slower than a helper 200 m away with a clear walking route).

$$\text{Rescue Priority Score} = 0.40 \times \text{estimated time-to-arrival factor} + 0.30 \times \text{distance factor} + 0.20 \times \text{Trust Score} + 0.10 \times \text{current availability factor}$$

Worked Example

For a fall incident on a downtown street, the system identifies two verified helpers within range:

Candidate	Distance	Estimated Time to Arrival	Trust Score	Available	Rescue Priority Score
Helper 1	120 m	2 minutes (clear walking route)	96 / 100	Yes	95%
Helper 2	300 m	5 minutes	85 / 100	Yes	78%

Helper 1 is dispatched first on the strength of a shorter estimated time to arrival and a higher Trust Score; Helper 2 remains on standby and is dispatched automatically if Helper 1 does not accept within the search window.

8.5 Staged Privacy Disclosure

Because a Nearby Helper is, by definition, an unrelated third party rather than a Guardian, disclosure of the affected user's information is staged and strictly limited at each step. Table 44 specifies exactly what is visible to a candidate helper before and after they accept a dispatch request.

Information	Before Acceptance	After Acceptance
Emergency type (e.g., possible fall)	Visible	Visible
Approximate distance and direction	Visible	Visible
Estimated time to arrival	Visible	Visible
Precise live location	Hidden	Visible, with continuous updates
User's name	Hidden	Hidden unless the user's settings permit disclosure
User's photograph	Hidden	Hidden unless the user's settings permit disclosure
Medical information	Hidden	Visible only to a verified Medical Helper (Section 8.6)

Table 44. Staged information disclosure to a candidate Nearby Helper, before and after they accept a dispatch request.

8.6 In-Field Coordination and Helper Permission Tiers

Once a helper accepts, the app opens Emergency Navigation Mode, presenting a turn-by-turn route to the incident location together with the latest known victim status (for example, "Fall detected, no response"). The Guardian and the affected user's own device receive periodic arrival-progress updates (for example, "Helper is approaching, ETA 45 seconds") for the duration of the approach. On arrival, the helper confirms with an explicit "I Have Reached the Location" action, which opens Emergency Assistance Mode and logs the confirmed arrival time into the incident record described in Section 7.6.

Tier	Eligibility	Permitted Actions
Basic Helper	Any verified, opted-in user	Navigate to the location; contact emergency services; confirm the victim's basic status
Medical Helper	Verified medical professional, paramedic, or certified first-aid responder	All Basic Helper actions, plus access to live heart rate, SpO ₂ , and consciousness-relevant status to inform first aid

Table 45. Nearby Helper permission tiers and the actions each is authorized to take.

8.7 Multi-Helper Task Coordination

When more than one verified helper responds to the same incident, the Emergency Orchestrator (Section 7) does not simply dispatch all of them to the same task. Instead, it distributes distinct roles automatically, mirroring how a professional first-response team divides labor at an incident scene.

Responder	Assigned Role
Helper A (fastest ETA)	Proceed directly to the victim's location
Helper B	Contact emergency services with the incident location and available context
Helper C	Wait at the nearest accessible point to receive and direct an arriving ambulance

Table 46. Representative task distribution when multiple Nearby Helpers respond to the same incident.

This coordination logic is what elevates the feature from a simple proximity alert into what the underlying design material terms an *AI Emergency Coordinator*: rather than only notifying nearby users that help is needed, the system actively manages who does what, reducing the chance that several well-intentioned responders duplicate the same action while no one performs another necessary one.

8.8 Fallback to Emergency Services

If no candidate helper accepts a dispatch request within a bounded search window (typically 30 seconds), the system does not continue searching indefinitely; it proceeds immediately to emergency-service escalation as described in Section 7.1 (Level 5), so that Nearby Rescue Network activation never adds latency to the professional-response path — it only ever runs in addition to it, never instead of it.

8.9 Summary Architecture

SENTRIX CORE → AI Detection → Risk Engine (E9) → parallel escalation channels:

- Guardian Network*
- Nearby Rescue Network (Sections 7.4, 14)*
- Emergency Services*

→ Emergency Coordination → Rescue Completed → Incident Report (Section 7.6)

Taken together with the severity-level model of Section 7.1 and the Guardian ecosystem of Section 12 (Cluster K), the Nearby Rescue Network completes SENTRIX CORE's positioning — argued in Section 3.1 — as a platform that coordinates a full emergency response rather than one that only issues an alert: combining artificial intelligence, verified community responders, the Guardian/family layer, and professional emergency services into a single coordinated escalation path.

9. Illustrative Operational Scenarios

This section grounds the architecture described in Sections 5 through 7 in concrete operational narratives. Twenty scenarios were developed to exercise every account type (Personal, Child, Elderly, Medical, Women Safety, Athlete), every operating mode (Phone-Only, Full Wearable, offline, low-battery), and every severity level defined in Section 7.1. Table 37 summarizes all twenty; Section 9.1 then presents five representative cases in full detail, including sensor activation, per-engine AI outputs, the Risk Score Engine (E9) calculation, and the resulting escalation timeline.

#	Scenario	User / Account	Key AI Engines Involved
1	Child leaves school safe zone	Child (8), standalone band	E10 GPS AI, E7 Digital Twin, E2 Activity AI, E9 Risk Engine
2	Child kidnapping attempt	Child (10), standalone band	E1 Fall AI, E6 Attack AI, E4 Distress Audio, E7 Digital Twin
3	Elderly fall at home	Elderly (78F), phone + band	E1b Pre-Fall AI, E1 Fall AI, E3 Health AI, E8 Context Engine
4	Heart attack during exercise	Medical (52M, CAD), phone + band	E3 Health AI, E2 Activity AI, E9 Risk Engine (medical override)
5	Seizure during sleep	Medical (34F, epilepsy), phone + band	E1 Fall AI (seizure pattern), E3 Health AI, E8 Context Engine
6	Panic attack in a public place	Adult (29M), phone only	E3 Health AI (phone fallback), E2 Activity AI, E8 Context Engine
7	Domestic violence — cannot call for help	Women Safety (38F), phone + band	E5 Panic Voice AI, E6 Attack AI, Silent Emergency Mode
8	Followed while walking alone at night	Women Safety (26F), phone + band	E10 GPS AI, E7 Digital Twin, E11 Predictive AI
9	Car accident — predicted, then confirmed	Adult (41M), phone + band	E11 Predictive AI, E1 Impact AI, E3 Health AI
10	Alzheimer wandering episode	Alzheimer protocol (81M), standalone band	E7 Digital Twin, E10 GPS AI, E8 Context Engine
11	Remote fall with network outage	Adult (31F), phone + band, hiking	E1 Fall AI, E3 Health AI, Offline Mode
12	Low-battery emergency, child band	Child (6), standalone band at 4%	E10 GPS AI, Battery Override Protocol
13	Athlete collapse during training	Athlete (24M), phone + band	Activity AI, Health AI, Predictive AI
14	Sleep emergency — cardiac	Medical (67F), phone + band	Sleep Monitor, Health AI, Silent Escalation
15	Post-surgical deterioration	Medical (55M), phone + band	Health AI, Continuous Monitor, Medical Packet
16	Elderly wandering at night	Elderly (83F), phone + band	Digital Twin, GPS AI, Alzheimer Protocol
17	Alzheimer patient lost in city	Alzheimer protocol (70M), standalone band	GPS AI, Behavioral AI, Emergency Services
18	Distress scream detected (bystander)	Adult (32F), phone + band	Audio AI, Context Engine, Nearby Users
19	Safe word triggered — coercion	Women Safety (45F), phone + band	Panic Voice AI, Override, Silent Escalation
20	Pre-fall prediction prevents injury	Elderly (74M), phone + band	Pre-Fall AI, Context Engine, Prevention

Table 37. Summary of the twenty illustrative operational scenarios developed to validate the SENTRIX CORE architecture across account types, operating modes, and severity levels.

9.1 Detailed Case Studies

The five cases below were selected to cover the architecture's most distinctive capabilities: covert threat detection for a child, predictive fall prevention for an elderly user, a medical override for a cardiac account, voice-based coercion detection for a Women Safety account, and pre-event accident prediction. Each case follows the same structure: narrative, sensor activation, the AI engines invoked, the Risk Score Engine (E9) calculation, the escalation timeline, and the final outcome.

Case 2 Child Kidnapping Attempt	
User	Layla, age 10 — Child account, standalone LTE band, no phone
Setting	Usual 800 m walking route home from school, 16:05
Narrative	A stranger in a car follows Layla and attempts to grab her arm. She cannot call for help while her attacker is present. The band operates entirely covertly.

Sensor Activation

- IMU: a violent acceleration spike (4.2g jerk) consistent with an arm grab, with rapid directional changes.
- PPG: heart rate rising from 92 to 163 BPM within 90 seconds.
- GPS: a route deviation — a stationary period followed by sudden movement in an unknown direction.
- Microphone: a 5-second audio burst triggered by the motion spike.

AI Engines Activated

Engine	Output
E1 Fall AI	Impact confidence 0.71 — violent motion, not a standard fall pattern
E6 Attack/Kidnapping AI	Motion (0.89) + panic HR (0.92) + GPS deviation (0.85) + band-only child context → threat probability 0.94
E4 Distress Audio AI	Distress vocalization 0.78; distress score 0.81
E7 Digital Twin	Route deviation from the learned pattern — behavioral anomaly 0.91 (critical)

Risk Score Calculation (Engine E9)

Component	Score
Fall / impact (25% weight)	$0.71 \times 25 = 17.8$
Health (20% weight)	$0.92 \times 20 = 18.4$
Audio (20% weight)	$0.81 \times 20 = 16.2$
GPS (15% weight)	$0.85 \times 15 = 12.8$
Behavior (10% weight)	$0.91 \times 10 = 9.1$
Context factor (10% weight)	$0.95 \times 10 = 9.5$
Total	83.8 → Attack AI threat 0.94 overrides to SILENT_EMERGENCY

Because the Attack AI threat probability (0.94) exceeds the 0.60 override threshold, Silent Emergency Mode activates immediately — no confirmation prompt, vibration, or display change occurs.

Escalation Timeline

+0s	Motion spike detected; Attack AI begins multimodal analysis
+3s	Threat probability 0.94; Silent Emergency Mode activated with zero external signals
+4s	Covert LTE transmission begins 1-second GPS streaming to the cloud
+5s	5-second distress-audio burst captured as evidence
+6s	Mother's phone receives a silent push notification only: possible danger, live GPS active, do not call
+9s	Police-escalation-ready packet generated (GPS, motion profile, audio evidence, AI decision log)
+15s	Trusted nearby SENTRIX users within 500 m receive an area alert
+40s	Mother authorizes emergency services in the app
~4m 22s	Police locate the vehicle from the GPS trajectory; Layla is recovered safely

Outcome. The child was recovered safely. A complete evidence package — GPS timeline, audio clip, motion profile, and AI decision log — was preserved with SHA-256 integrity hashing for subsequent legal proceedings.

Case 3 Elderly Fall at Home, with Pre-Fall Prediction

User	Margaret, 78F — Elderly account, phone + band, lives alone
Setting	Home bathroom, 06:48
Narrative	Margaret slips on a wet tile, falls hard, and briefly loses consciousness. The Pre-Fall AI detected balance deterioration two seconds before impact; she cannot press any button after falling.

Sensor Activation

- IMU: a pre-fall instability window at t–2s, followed by a high-g impact (3.8g vertical deceleration).
- IMU: orientation change from vertical to horizontal, and 23 seconds of post-fall inactivity.
- PPG: heart rate present but mildly elevated (78 BPM) — consistent with shock, not a cardiac event.
- Microphone: an impact sound captured at the moment of the fall; no vocalization, since the user is unconscious.

AI Engines Activated

Engine	Output
E1b Pre-Fall AI	Lateral sway and 38% gait asymmetry at t–2s — pre-fall risk 0.73; a silent vibration alert is sent
E1 Fall AI	Impact confirmed at 3.8g; orientation change within 0.8s; fall probability 0.97
E3 Health AI	Heart rate stable (78 BPM), SpO ₂ 96% — no cardiac event
E8 Context Engine	06:48, home bathroom, elderly account → context factor 1.8 (elevated concern)

Risk Score Calculation (Engine E9)

Component	Score
Fall (25% weight)	$0.97 \times 25 = 24.3$
Health (20% weight)	$0.15 \times 20 = 3.0$
Audio (20% weight)	$0.52 \times 20 = 10.4$
GPS / Behavior	0.0 (at home, recognized safe zone)
Context factor (elderly, alone, bathroom)	×1.8 multiplier applied to the raw total
Total	37.7 raw → 82.3 after context adjustment → CRITICAL

Escalation Timeline

t-2s	Pre-Fall AI detects lateral sway; silent vibration sent
t+0s	Impact confirmed (0.97); inactivity timer starts
+5s	Elderly-account fall timeout (5s) expires with no response; CRITICAL activated automatically
+6s	LTE activated; home GPS location confirmed
+7s	Daughter notified by push and SMS: possible fall, no response, live location attached
+8s	Evidence collected: fall motion sequence, pre-fall trajectory, impact audio, health log
+10s	Trusted nearby users receive an area alert
~1 min	Daughter authorizes emergency services in the app; ambulance dispatched with fall context
~4 min	Ambulance arrives; Margaret is conscious but disoriented, with a wrist fracture

Outcome. Margaret survived with a manageable wrist fracture. For an elderly person living alone, the typical time to discovery after a fall without such a system is measured in hours; reducing that interval to minutes is presented as the core life-safety value of SENTRIX CORE for this user group.

Case 4 Heart Attack with Medical Override

User	Robert, 52M — Medical account, known coronary artery disease, phone + band
Setting	Morning walk in a park, 07:23
Narrative	Robert begins experiencing chest tightness during a walk. The Health AI detects progressive heart-rate, HRV, and SpO ₂ changes before he recognizes the severity himself, and — critically — correctly overrides his own reflexive “I’m okay” response.

Sensor Activation

- PPG: heart rate rising from 88 to 164 BPM over 8 minutes; HRV (RMSSD) collapsing from 45 ms to 11 ms; SpO₂ falling from 97% to 90%.
- IMU: walking pace slowing from 4.2 km/h to a full stop.
- GPS: stationary in a recognized, low-risk park location.

AI Engines Activated

Engine	Output
E3 Health AI (band rule layer)	HR > 160 BPM and SpO ₂ < 92% thresholds crossed; immediate band-level alert
E3 Personal Autoencoder	Reconstruction error 4.2× baseline; pattern matches a cardiac-anomaly template; health risk 0.94
E3 Population Model (XGBoost)	Age, sex, and CAD history combined with the HR/HRV/SpO ₂ pattern → cardiac-event probability 0.87
E2 Activity AI	Detects the transition from walking to stationary as a distress indicator

Risk Score Calculation (Engine E9, Medical Account)

Component	Score
Health (20% weight, medical accounts apply a 1.5× boost)	0.94 × 20 = 18.8 → 28.2
Behavior (pace deceleration to a stop)	0.45 × 10 = 4.5
Context factor (medical account, alone, stationary)	1.6 × 10 = 16.0
Total	49.5 base → 94 after the medical-account multiplier → CRITICAL

Medical accounts use a 3-second confirmation timeout. Robert, conscious but distressed, taps the logo once out of reflex — the standard “I’m okay” gesture. Because his Health Risk score (0.94) exceeds the 0.90 medical override threshold, this single-tap response is deliberately ignored and escalation continues — a bypass designed specifically for medical accounts, where users may understate the severity of what they are experiencing.

Escalation Timeline

+0s	HR reaches 138 BPM; autoencoder reconstruction error 2.1× baseline; Warning state
+45s	HR 164 BPM, SpO ₂ 90%, HRV 11 ms; Risk Score 94; CRITICAL medical protocol
+48s	3-second medical timeout begins; band vibration and notification sent
+51s	Robert taps once ('I'm okay'); overridden by the 0.94 health-risk score; escalation continues
+52s	LTE activated; GPS location streamed
+53s	Emergency packet generated with medical profile: CAD history, medications, baseline vitals
+54s	Wife notified by push and SMS with live GPS: possible cardiac event
~1 min	Wife authorizes emergency services (pre-authorized in account settings)
~5 min	Ambulance arrives with the pre-transmitted cardiac history; appropriate medication given immediately

Outcome. Robert survived an ST-elevation myocardial infarction. His cardiologist credited the early arrival of emergency medical services — enabled by the automatically generated, medical-context emergency packet — with preventing permanent cardiac damage.

Case 19 Safe Word Override — Coercion Scenario

User	Amira, 45F — Women Safety Mode, phone + band
Setting	Shopping centre, 15:30
Narrative	An unknown man appears to coerce Amira into walking with him — a possible abduction setup. She calmly says 'I'm okay, I just need to go,' but her voice tells a different story.

Sensor Activation

- Microphone: a 3-second voice sample captured during the confirmation prompt.
- PPG: heart rate spiking from 76 to 131 BPM.
- IMU: walking pace and direction altered in a pattern consistent with external coercion.

AI Engines Activated

Engine	Output
E3 Health AI	Sudden HR spike — stress pattern; health risk 0.81
E10 GPS AI	Direction change from the intended route — forced-movement pattern; GPS risk 0.72
E5 Panic Voice AI	Amira says 'I'm okay'; acoustic analysis shows fear 0.89, pitch deviation 2.4× baseline, shimmer 1.9× baseline — override flag set, verbal claim rejected
E6 Attack/Kidnapping AI	Combining HR, forced movement, and the voice override → threat probability 0.83

The Panic Voice override (E5) triggers Silent Emergency Mode directly: Amira's verbal claim of safety is rejected on acoustic grounds, and the system continues escalating while outwardly appearing to accept her response — no visible signal is shown on the device.

Escalation Timeline

+0s	HR spike and GPS direction change detected; Attack AI begins analysis
+5s	Confirmation prompt sent; Amira responds verbally 'I'm okay'
+7s	Voice analysis: fear 0.89; override triggered; verbal claim rejected
+8s	Silent Emergency Mode activated; no visible signal on the device
+9s	Sister notified by silent push: possible coercion, live GPS active
+10s	Covert GPS streaming begins at 5-second intervals; a 5-second audio evidence sample is captured
~1 min	Amira steps into a shop; the coercer does not follow into the crowded space
~1.5 min	Amira triple-taps to cancel SOS and confirms safety via biometric check; sister authorizes a police report

Outcome. Amira escaped safely. A police report was filed with the GPS trace of the encounter, and the individual involved was later identified in connection with a pattern of similar incidents in the area. This case demonstrates the Panic Voice override functioning exactly as designed: detecting coercion even when the user verbally denies danger.

Case 9 Car Accident — Predicted, Then Confirmed

User	Ahmed, 41M — Adult account, phone + band
Setting	Highway commute, 08:25–08:45
Narrative	Fatigue sets in after a poor night's sleep. The Predictive Incident AI (E11) detects a rising accident probability twenty minutes before a collision occurs; the driver dismisses the warning and, at 08:45, a moment of microsleep causes him to swerve and strike a barrier.

Pre-Accident Sensor Activation (08:25)

- PPG: heart rate gradually rising (72 → 88 → 102 BPM) with declining HRV — a fatigue signature.
- IMU: vehicle-motion signature classified as Driving, with subtle irregularities emerging in vehicle handling.

Predictive AI Output

Engine E11 combines the rising heart-rate/HRV fatigue trend with the vehicle-motion irregularity into an accident probability of 0.71 and issues a proactive in-app warning (“Driving fatigue detected — consider stopping”), placing the system into Readiness Mode: GPS pre-streaming begins and the LTE modem moves to standby, twenty minutes before any impact occurs.

Post-Impact Sensor Activation (08:45)

- IMU: an 8.4g deceleration spike consistent with a barrier impact, followed by a sudden transition to stationary.
- PPG: heart rate spiking from 88 to 158 BPM in the immediate post-impact stress response.
- GPS: stationary on the highway shoulder.

Risk Score Calculation (Engine E9)

Component	Score
Impact (25% weight)	$0.94 \times 25 = 23.5$
Health (20% weight)	$0.78 \times 20 = 15.6$
GPS (15% weight)	$0.65 \times 15 = 9.8$
Behavior (10% weight)	$0.55 \times 10 = 5.5$
Context factor (highway, driving, sudden stop)	$1.4 \times 10 = 14.0$
Total	68.4 base → 91 after the driving-accident context boost → CRITICAL

Escalation Timeline

08:25	Predictive AI: accident probability 0.71; proactive warning issued and dismissed
08:45	Impact detected at 8.4g; Fall/Impact AI confidence 0.94
+10s	Automotive confirmation timeout (10s) expires with no response; CRITICAL activated
+11s	LTE and GPS streaming begin; emergency packet includes the preceding 20-minute fatigue log
+12s	Wife notified by push and SMS with live GPS: possible accident on the highway
+13s	Emergency services dispatched with accident context and highway coordinates
~5 min	Police and ambulance arrive; Ahmed sustains only minor whiplash and airbag bruising

Outcome. Ahmed recovered from minor injuries. Beyond the immediate response, the twenty-minute predictive fatigue warning that preceded the collision is highlighted as the highest-value category of feature in the SENTRIX CORE architecture: an intervention that could, in other instances, prevent the accident from occurring at all rather than merely accelerating the response to it.

10. Patent Potential Analysis

The wearable device itself is unlikely to be independently patentable, since smart bands already exist as a well-established product category. However, several sub-systems represent plausible patent opportunities.

Patent Area 1 — Multi-Modal AI Emergency Decision System

A method combining physiological signals, motion data, behavioral patterns, and location context to generate a personalized emergency probability through weighted, multi-engine fusion.

Patent Area 2 — Adaptive Personal Safety Digital Twin

A system that learns individual behavior, updates personal baselines, and automatically adjusts detection thresholds through continuous self-supervised learning on the user's own sensor data.

Patent Area 3 — Autonomous Emergency Escalation Engine

A decision system that determines whether to request user confirmation, contact guardians, or escalate to emergency services, based on multi-modal confidence scoring, including a voice-override capability that can detect coercion despite a verbal denial from the user.

11. Research Novelty and State-of-the-Art Comparison

Research Novelty Statement

"This work proposes an autonomous AI-powered wearable safety ecosystem integrating multi-modal sensor fusion, personalized Digital Twin modeling, edge intelligence, and predictive emergency decision-making to overcome the limitations of existing reactive wearable safety systems."

11.1 Comparison With State-of-the-Art

Feature	Existing Wearables	Research Systems	SENTRIX CORE
Health Monitoring	✓	✓	✓
Fall Detection	✓	✓	✓
AI Prediction	Limited	Partial	✓
Digital Twin	Rare	Research	✓
Sensor Fusion	Limited	Research	✓
Edge AI	Partial	Research	✓
Autonomous Emergency	Limited	Rare	✓
Guardian Ecosystem	Partial	Rare	✓
Continuous Learning	Rare	Research	✓
Standalone LTE Safety	Some	Rare	✓

Table 4. SENTRIX CORE against existing wearables and published research systems.

11.2 Final Research Position

Based on the reviewed academic and industrial landscape, SENTRIX CORE is not positioned as another smartwatch. The project's position is best summarized as an **AI safety-intelligence platform embedded into wearable technology**. The closest existing technologies each solve one part of the problem: Apple addresses health monitoring and reactive emergency alerting; Empatica addresses research-grade physiological sensing; Medical Guardian addresses professional emergency response; the Digital Twin research literature addresses personalization; and TinyML addresses edge intelligence. SENTRIX CORE's contribution is to combine these previously separate capabilities into a single autonomous safety ecosystem.

12. Application Architecture and Screen Taxonomy

Sections 5 through 7 described SENTRIX CORE's AI decision layer in isolation. This section describes how that decision layer is exposed to end users through the companion mobile application, which we treat as a first-class research artifact in its own right: it is the surface through which the Digital Twin, the fusion engines, and the autonomous emergency escalation logic actually reach a person. The application is organized as a single codebase covering seventy-four screens, grouped into fifteen functional clusters, and governed by two operating modes (Phone-Only and Full Wearable) rather than as two separate products — a design choice discussed further in Section 12.3.

12.1 Design Philosophy

Three principles guided the screen taxonomy. First, every screen maps to one or more of the thirteen AI engines (E1–E13) described in Section 6, or to a supporting account, guardian, or system-management function; no screen exists purely for navigational decoration. Second, overlapping or near-duplicate screens observed in early iterations of the design (for example, separate “Edit Profile,” “Medical Conditions,” and “Cloud Sync” screens) were consolidated into parent screens such as Profile, Health Profile, and Privacy & Cloud, which keeps the total screen count manageable without removing functionality. Third, rather than shipping a reduced “lite” app for users without a paired band and a full app for those with one, SENTRIX CORE uses a single application that adapts its available features to the current operating mode, discussed in Section 12.3.

12.2 Screen Inventory by Functional Cluster

Table 5 summarizes cluster-level counts; the full screen-by-screen inventory follows in Tables 6–19. Screens 66–69 are four screens added after an initial 65-screen design pass specifically to raise the application to an enterprise-grade standard; they are listed within their logical cluster below rather than as a separate appendix, since each maps directly onto an existing engine or subsystem.

A subsequent revision added Screens 70–74 - a conversational AI assistant, a live community incident map, band personalization controls, a gamified safety-engagement layer, and a public-facing product information screen. Each is listed within its logical cluster below, except the engagement screen, which forms a new Cluster O.

Cluster	Name	Screen #s	Count
A	Onboarding & Account Setup	1–10	10
B	Home & Profile Management	11–19	9
C	Smart Band Management	20–22, 72	4
D	AI Intelligence Center	23–27, 67–68, 70	8
E	Health Monitoring	28–33	6
F	Location & Tracking	34–37, 71	5
G	Child Mode	38–41	4
H	Elderly Mode	42–45	4
I	Women Safety Mode	46–49	4
J	Emergency & Incident Subsystem	50–55, 66	7
K	Guardian Subsystem	56–61	6
L	Notifications	62	1
M	Reports	63	1
N	System & Security	64–65, 69, 74	4
O	Engagement & Gamification	73	1

Table 5. Screen clusters and counts, totaling seventy-four screens.

Cluster A — Onboarding & Account Setup (Screens 1–10)

Covers the full first-run journey from initial launch to a fully calibrated, permissioned account, ending with the Home Dashboard.

#	Screen	Function
1	Splash Screen	Branded loading screen shown while core services initialize and any existing session is checked.
2	Onboarding	A short introductory carousel explaining SENTRIX CORE's value proposition — predictive safety, the Digital Twin, and autonomous emergency response — to first-time users.
3	Login	Authenticates a returning user.
4	Sign Up	Collects the information required to create a new account.
5	OTP Verification	Verifies the user's phone number or email via one-time password before account activation.
6	Choose Account Type	Selects the profile governing which dashboards and features are surfaced: Personal, Child, Elderly, or Medical Guardian.
7	Health Information	Collects baseline health data used to seed the Health Anomaly AI (E3) and Digital Twin (E7) baselines.
8	AI Calibration	Runs the initial calibration period during which the Digital Twin begins learning the user's personal motion, heart-rate, and behavioral baselines.
9	Pair Smart Band	Guides Bluetooth/BLE pairing with the wearable band and determines whether the app subsequently runs in Full Wearable Mode or Phone-Only Mode.
10	Permission Center	Presents and manages every system permission the app needs — GPS, Bluetooth, Microphone, Camera, SMS, Phone, Notifications, Battery Optimization — with a plain-language rationale for each.

Cluster B — Home & Profile Management (Screens 11–19)

Covers the application's central hub and the account-level settings a user configures infrequently rather than daily.

#	Screen	Function
11	Home Dashboard	The application's central hub: current risk status, quick access to Health, AI, Tracking, Emergency, and Guardian modules, and the active operating-mode banner.
12	Profile	Manages personal information, profile picture, and account edits.
13	Health Profile	Stores height, weight, blood type, allergies, medical conditions, and medications, used by the Health Anomaly AI (E3) and included in the emergency data packet described in Section 6.
14	Emergency Contacts	Manages the people notified during an escalated emergency.
15	Guardian Management	Manages which guardians are linked to the account (see also the dedicated Guardian subsystem, Cluster K).
16	Linked Devices	Lists every device — band, phone, tablet — associated with the account.
17	Subscription	Manages the user's SENTRIX CORE plan and billing.
18	Settings	Consolidates theme, language, notification settings, quiet hours, and alert customization.
19	Privacy & Cloud	Consolidates privacy controls, data permissions, cloud synchronization, backup, and encryption status.

Cluster C - Smart Band Management (Screens 20-22, 72)

Covers the operational lifecycle of the paired wearable band, active only in Full Wearable Mode.

#	Screen	Function
20	Smart Band Status	Displays connection status, battery, LTE/eSIM state, GPS status, and on-device diagnostics.
21	Firmware Update	Manages over-the-air firmware updates, working in concert with the AI Governance / Model Health Engine (E13).
22	Sensor Calibration	Runs and reviews sensor-level calibration (IMU, PPG, microphone) that feeds the Sensor Reliability AI (E12).
72	Band Customization	Configures the band's vibration patterns, display brightness, and quick-action shortcuts for Full Wearable Mode.

Cluster D - AI Intelligence Center (Screens 23-27, 67-68, 70)

Exposes the outputs of the thirteen-engine AI architecture from Section 6 directly to the user, including the two explainability screens added to raise user trust in autonomous decisions.

#	Screen	Function
23	AI Dashboard	The Risk Engine's live output: overall AI confidence, system status, and the current risk score from the Risk Score Engine (E9).
24	AI Analysis	Surfaces the individual outputs of the Context Engine (E8), Attack/Kidnapping Detection (E6), Fall Detection (E1/E1b), and Health Analysis (E3) in one view.
25	Digital Twin	Visualizes the user's personal behavioral model (E7): learned routines, location clusters, and current deviation from baseline.
26	Risk Timeline	Plots the Risk Score Engine's (E9) output over time so the user or guardian can review recent risk trends.
27	AI Training Status	Reports on-device and cloud training progress, the date of the last model update, current accuracy, and which of the thirteen engines are actively running.
67	AI Explainability (XAI)	Answers "why did the AI make this decision?" by decomposing a given risk score into its contributing engines and weights (e.g., Fall 40%, Heart Rate 25%, Audio 20%, GPS 15%) — directly supporting user trust in the autonomous decision layer of Section 5.
68	AI Model Status	Reports the deployed model version, last update date, accuracy, and active-model list, complementing the AI Governance / Model Health Engine (E13).
70	AI Chat Assistant	Conversational assistant that explains AI alerts in plain language and answers user questions about the current risk score and recent engine decisions.

Cluster E - Health Monitoring (Screens 28-33)

Presents the physiological signal streams consumed by the Health Anomaly AI (E3).

#	Screen	Function
28	Live Health	Real-time heart rate, HRV, and SpO ₂ .
29	Heart Analytics	Detailed heart-rate and HRV trends and history.
30	Stress Analytics	Stress-level trends derived from HRV and behavioral signals.
31	Sleep Analytics	Sleep-stage and sleep-quality analysis derived from motion and heart rate during rest.
32	SpO ₂ Analytics	Blood-oxygen saturation trends and history.
33	Health History	A consolidated, exportable history of all physiological metrics.

Cluster F - Location & Tracking (Screens 34-37, 71)

Presents the outputs of the GPS + Safe Zone AI (E10) and the location component of the Digital Twin (E7).

#	Screen	Function
34	Live Tracking	Real-time map view of the user's current location.
35	Route History	Historical GPS route log.
36	Safe Zones	Defines and manages geofenced areas treated as low-risk by E7 and E10.
37	Unsafe Zones	Defines and manages geofenced areas that raise the Context Engine's (E8) risk multiplier when entered.
71	Live Incident Map	Displays nearby active alerts and incidents shared by the Nearby Rescue Network (Section 8) on a live map.

Cluster G — Child Mode (Screens 38–41)

Replaces the standard Home Dashboard and related screens when the Child account type (Screen 6) is active, oriented toward guardian oversight rather than self-monitoring.

#	Screen	Function
38	Child Dashboard	A simplified, guardian-facing view of the child's current status.
39	School Monitoring	Tracks arrival at and departure from school-designated safe zones.
40	Child Activity	Summarizes the child's motion and activity patterns for the day.
41	Child Route	Displays the child's traveled route for guardian review.

Cluster H — Elderly Mode (Screens 42–45)

Emphasizes fall risk, wandering, and medication adherence, drawing directly on the Digital Twin wandering-detection example described in Section 6 (Engine E7).

#	Screen	Function
42	Elderly Dashboard	A guardian-facing view emphasizing fall risk, wandering, and medication status.
43	Wandering Detection	Surfaces Digital Twin (E7) behavioral-deviation alerts for unexpected nighttime or unfamiliar-route movement.
44	Medication	Manages medication schedules and adherence reminders.
45	Recovery Route	Displays a suggested route back to a known safe location once wandering is detected.

Cluster I — Women Safety Mode (Screens 46–49)

Built around the Attack/Kidnapping Detection engine (E6) and its Silent Emergency workflow.

#	Screen	Function
46	Women Safety Dashboard	A dashboard emphasizing threat detection (E6) and the Silent Emergency workflow.
47	Silent Emergency	Activates and manages Silent Emergency Mode: covert GPS streaming, guardian alerting, and evidence capture without an audible or visible alarm, as described for Engine E6.
48	Safe Route	Suggests routes that avoid flagged unsafe zones, informed by the GPS + Safe Zone AI (E10).
49	Trusted Nearby Users	Shows other trusted app users currently nearby who can be alerted in an emergency.

Cluster J — Emergency & Incident Subsystem (Screens 50–55, 66)

The application's most safety-critical cluster, spanning an emergency in progress through to a fully documented, replayable post-incident record.

#	Screen	Function
50	Emergency Screen	The active-emergency interface: countdown timer, cancel button, dispatch progress, and current risk score from the Risk Score Engine (E9).
51	Emergency History	A log of all past emergency events.
52	Evidence Center	Consolidates audio, camera, and video evidence captured during an emergency (see Distress Audio AI, E4).
53	Emergency Report	Generates a formal, shareable report of a given emergency for guardians, medical staff, or authorities.
54	Incident Replay	Reconstructs the full event timeline: walking, heart-rate change, audio, fall, AI decision, countdown, SOS, guardian notification, live tracking, and emergency closure.
55	Emergency Simulation	Lets the user rehearse the emergency workflow end-to-end without dispatching a real alert.
66	Incident Summary	A human-readable post-incident report combining a status header, a timestamped timeline, a health snapshot, and a list of actions taken — detailed in Section 12.4.

Cluster K — Guardian Subsystem (Screens 56–61)

The caregiver-facing counterpart to Clusters G, H, and I, allowing one account to monitor multiple dependents.

#	Screen	Function
56	Guardian Dashboard	A guardian's overview of every linked member's current status and risk level.
57	Member Details	Detailed profile and live status for one linked member.
58	Add Member	Invites and links a new dependent or family member.
59	Verification	Confirms guardian identity and relationship before granting access.
60	Permissions	Configures exactly what each guardian can see or do.
61	Guardian Alerts	The guardian-facing alert feed for linked members' risk events.

Cluster L — Notifications (Screen 62)

#	Screen	Function
62	Notification Center	Consolidates notification history, alert customization, and quiet-hours configuration.

Cluster M — Reports (Screen 63)

#	Screen	Function
63	Reports Dashboard	Tabbed access to Weekly, Monthly, AI, Health, and Safety reports.

Cluster N - System & Security (Screens 64-65, 69, 74)

Operational and security transparency screens that support the Sensor Reliability AI (E12) and AI Governance Engine (E13) at the application layer.

#	Screen	Function
64	System Health	Live status of GPS, Bluetooth, internet, AI services, cloud connectivity, the smart band, and sensors.
65	Offline Mode	Shows which features remain available offline, last sync time, pending uploads, and local storage status.
69	Account & Device Security	Manages active sessions, trusted devices, login history, multi-factor authentication, and security alerts.
74	About SENTRIX CORE	Presents the product vision, team, version and build information, and links to legal and support resources.

Cluster O - Engagement & Gamification (Screen 73)

A lightweight motivational layer that encourages consistent safety-relevant engagement without gating any core safety function behind it.

#	Screen	Function
73	Achievements & Safety Score	Gamified layer of badges, streaks, and a composite safety score that rewards consistent monitoring, calibration, and safe behavior over time.

12.3 Operating Modes: Phone-Only vs. Full Wearable

Rather than shipping two separate applications, SENTRIX CORE ships one application whose available features are governed by a runtime operating mode. Phone-Only Mode is active whenever no smart band is paired; Full Wearable Mode activates automatically once pairing (Screen 9) succeeds. A persistent banner at the top of the Home Dashboard (Screen 11) always communicates the active mode — “Phone-Only Protection” or “Band Connected — Protection Level: Advanced” — so the user always knows which capabilities are currently available.

Capability	Phone-Only Mode	Full Wearable Mode
SOS / Emergency Escalation	✓	✓
GPS / Live Tracking / Safe Zones	✓	✓
AI Risk Engine (E9)	✓	✓
Distress Audio AI (E4)	✓	✓
Attack / Kidnapping Detection (E6)	✓	✓
Safe Word / Panic Voice AI (E5)	✓	✓
Digital Twin (E7)	✓	✓
Guardian System	✓	✓
Camera / Audio Evidence	✓	✓
Heart Rate / SpO ₂ / HRV (E3)	✗ (no PPG sensor)	✓
TinyML Fall AI (E1 / E1b)	✗ (phone IMU only, reduced fidelity)	✓ (dedicated on-band IMU)
Standalone LTE / eSIM	✗	✓
Band Battery / Diagnostics	✗	✓
Firmware Update / Sensor Calibration	✗	✓
Independent SOS Without Phone	✗	✓
Band Confirmation Button / Vibration Alerts	✗	✓

Table 20. Feature availability by operating mode. Core safety functions (SOS, GPS, the AI Risk Engine, and the Guardian system) remain fully active in Phone-Only Mode; only band-dependent physiological and standalone-connectivity features are affected.

This design keeps the five research gaps identified in Section 4 addressed even for users who have not yet paired a band — in particular, Gap 4 (smartphone dependency) is only partially mitigated in Phone-Only Mode, since standalone LTE requires the band, but Gaps 1–3 and 5 (unified architecture, prediction, personalization, and autonomous escalation) remain fully available through the phone's own sensors and the cloud-side Digital Twin.

12.4 Incident Summary and Incident Replay

Screens 54 (Incident Replay) and 66 (Incident Summary) together implement a design principle carried over from aviation and industrial incident reporting: after any Fall, Attack, or Medical Emergency event, the user or guardian should see the reasoning behind the system's decisions, not merely a terse “emergency alert was sent” notification. Incident Summary presents this as a structured, four-part record.

Part 1 — Status header.

A compact header identifying the incident number, date, time, final status, and the Risk Score Engine's (E9) final output for the event — for example, Incident #2458, 22 May 2026, 8:34 PM, Status: Emergency Completed, Risk Score: 94/100.

Part 2 — Timeline.

8:34:01 PM — Walking normally → 8:34:08 — Heart rate rises to 145 BPM → 8:34:10 — Distress audio detected (E4) → 8:34:12 — Sudden impact → 8:34:13 — Fall detected (E1) → 8:34:15 — Countdown started → 8:34:25 — No response → Emergency activated → Guardian notified → Live tracking started → Evidence uploaded → Emergency closed.

This timeline is the human-readable rendering of exactly the same sequence of engine outputs described abstractly in the Engine E1 and E9 examples in Section 6, making Incident Summary the user-facing counterpart of the underlying AI decision log.

Part 3 — Health snapshot.

A chart of heart rate across the incident window (for example, 82 → 148 → 160 → 132 → 95 BPM), giving the user or a reviewing clinician a compact physiological record of the event alongside the behavioral timeline.

Part 4 — Actions taken.

- SMS sent to emergency contacts
- Call placed to emergency contacts
- Guardian notified
- Live tracking enabled
- Cloud backup of evidence saved

Because Incident Summary and Incident Replay share the same underlying event log, a future version could support exporting either view directly into the Emergency Report (Screen 53) or attaching it to the AI Explainability screen (Screen 67) for a combined “what happened and why the AI acted” record.

12.5 Navigation Flow

The onboarding sequence (Cluster A) is strictly linear up to AI Calibration, after which the flow branches on a single decision — whether a smart band is paired — before both branches rejoin at the Home Dashboard, from which every other cluster is reachable.

*Splash → Onboarding → Login / Sign Up → OTP Verification → Choose Account Type → Health Information → AI Calibration → **Pair Smart Band?***

- Yes → Full Wearable Mode*
- No → Phone-Only Mode*

Both branches → Home Dashboard → { Health | AI | Emergency | Guardian } → { Tracking | Reports | Evidence | Settings }

From the Home Dashboard, the four primary clusters — Health, AI, Emergency, and Guardian — are always one tap away, while Tracking, Reports, Evidence, and Settings are reachable as a second tier beneath them. This shallow hierarchy keeps the safety-critical Emergency Screen (Screen 50) within two taps of any point in the application at all times.

12.6 UI State Design System

Rather than adding further screens, every screen in Clusters A-O is designed against five standard states, so that the same information architecture behaves predictably under real-world conditions such as sensor dropout, empty history, or lost connectivity. This is analogous to the Sensor Reliability AI (E12) and AI Governance Engine (E13) at the AI layer: just as those engines keep the decision layer honest about degraded input, the UI state system keeps the interface honest about degraded conditions rather than silently showing stale or misleading data.

State	Purpose	Representative Examples
Loading	Communicates that a screen is waiting on a sensor, the AI engines, or a network call, rather than appearing frozen.	"Loading Sensors...", "Loading AI...", "Loading GPS..."
Empty	Shown when a data set legitimately contains nothing yet, distinguishing this from an error.	"No Emergency History," "No Notifications," "No Safe Zones"
Error	Surfaces a specific, actionable failure rather than a generic error message.	"GPS Lost," "Band Disconnected," "Cloud Offline," "Sensor Error," "Permission Denied"
Success	Confirms that a user- or system-initiated action completed as expected.	"Band Connected," "Emergency Cancelled," "Cloud Synced," "AI Updated"
Offline	Distinguishes graceful degradation from failure when connectivity is lost, and shows what will happen once it returns.	"Offline Mode," "Cached Data," "Pending Upload," "Reconnect"

Table 21. The five standard UI states applied consistently across all seventy-four screens.

12.7 Multi-Platform Companion Ecosystem

The 74-screen mobile application is the primary and most thoroughly specified surface, but it is designed as one node in a wider ecosystem, consistent with how comparable platforms reviewed in Section 3 (Apple, Google) extend a single mobile experience across companion form factors. Table 22 summarizes the four companion surfaces identified for SENTRIX CORE.

Platform	Primary Users	Key Capabilities
Mobile App (Android / iOS)	Individual users, families	The full 74-screen experience described in Sections 9.1-9.6; the primary interface for Personal, Child, Elderly, and Women Safety modes.
Guardian Tablet App	Families, hospitals, care facilities	A larger-format view for monitoring multiple linked members at once: all dependents, large-format maps, health status, alerts, and multi-person monitoring.
Web Dashboard	Companies, hospitals, schools, security teams, families	Browser-based access to a live dashboard, reports, incident review, device management, over-the-air update management, and AI analytics.
Emergency Operator Dashboard	Emergency response centers	A responder-facing view of a live incident, GPS position, the user's medical profile, current risk score, captured evidence, and the incident timeline.
Admin Dashboard	SENTRIX CORE platform operators	Full system administration: user and guardian management, device fleet management, AI model versioning, OTA rollout, cloud infrastructure, analytics, and audit logs.

Table 22. The multi-platform SENTRIX CORE ecosystem, of which the mobile application described in this section is the primary and most detailed surface.

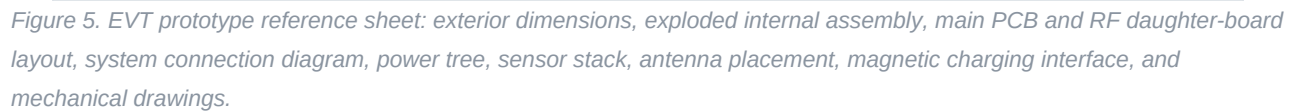
Taken together with the thirteen-engine AI architecture (Section 6), the seventy-four-screen mobile application, and the four companion platforms summarized above, SENTRIX CORE's specification spans the full stack expected of an enterprise-grade safety platform: sensing hardware, edge and cloud AI, a consumer application, caregiver and institutional tooling, and emergency-responder integration.

13. Hardware Architecture and Industrial Design

Sections 6 and 9 described SENTRIX CORE's AI decision layer and the mobile application that exposes it to users. This section specifies the physical wearable itself — the smart safety band whose sensors feed the thirteen-engine architecture of Section 6 and whose on-device compute realizes the “Band (edge)” inference tier referenced throughout that section. The specification below reflects an Engineering Validation Test (EVT) prototype: a design methodology consistent with commercial smart bands prior to mass production, built from industrial-grade, commercially available components sourced from established distributors (Mouser, DigiKey, LCSC).

13.1 Product Overview and Hardware Philosophy

SENTRIX CORE is best understood as a very small special-purpose computer built into a smartwatch-band form factor. Nine subsystems are orchestrated by a single central microcontroller: a user interface (touch button, RGB LEDs, vibration motor), a central processor, motion sensors, health sensors, GPS positioning, LTE cellular connectivity, battery and power management, memory and a secure element, and an on-device AI inference engine. Every subsystem exists to serve one architectural principle: sensors only collect raw data, and the microcontroller is solely responsible for reading that data, running AI inference, making contextual decisions, and managing communication — the same separation-of-concerns principle used throughout the software architecture in Sections 6 and 11.



13.2 System Architecture and Data Flow

The hardware follows a layered architecture that mirrors industrial embedded-systems practice: hardware abstraction, control, analysis, decision-making, and communication are kept as separate layers. Sensors collect raw motion, health, GPS, and environmental data only; the microcontroller manages all hardware and runs firmware; on-device AI engines analyze sensor data; the Risk Engine (E9, Section 6) determines the danger level; an Emergency Orchestrator decides what actions to take; a Communication Manager selects the best connectivity method; and the cloud and mobile application (Section 12) display information and coordinate with guardians.

Sensors → Drivers → Sensor Manager → AI Models → Fusion Engine → Risk Engine → Decision Engine → Emergency Orchestrator → Communication Manager → Cloud → Guardian → Mobile App

13.3 Core Processor (MCU)

The system is built around a single central processor, the Nordic nRF52840, which does not itself measure heart rate, determine location, or operate the LTE radio, but instead coordinates every subsystem that does.

CPU	ARM Cortex-M4F with hardware floating-point unit (FPU)
Clock speed	64 MHz
Flash memory	1 MB
RAM	256 KB
Wireless	Bluetooth Low Energy 5.3
Package	QFN73 (7 × 7 mm)

The nRF52840 integrates a full peripheral suite used across the design: timers for precise scheduling; GPIO for general-purpose control; SPI for high-speed serial communication (flash memory); I2C for the multi-device sensor bus; UART for the LTE and GPS modules; PWM for motor and LED control; an ADC for analog signals; the BLE radio itself; a hardware random-number generator for security operations; DMA for efficient data transfer; and a priority-based interrupt controller.

On power-on, the following boot sequence completes in under one second:

Battery → PMIC → 3.3V regulated rail → MCU boot → firmware starts → initialize drivers → initialize sensors → start BLE → start AI engine → enter idle loop

The MCU's six core responsibilities are: system boot and power management; reading all sensor data at configured intervals; running on-device AI inference; making contextual decisions; managing data transmission; and optimizing energy use across every subsystem.

13.4 Sensor Suite

Five sensor groups feed the AI engines described in Section 6. Each sensor communicates with the MCU over I2C, SPI, or I2S and does not process or transmit data autonomously.

Motion Sensor — Bosch BMI270

The BMI270 combines a 3-axis accelerometer and a 3-axis gyroscope in a single 2.5×3.0 mm package and is the primary input to the Fall Detection (E1/E1b), Human Activity Recognition (E2), and Attack Detection (E6) engines. It samples at up to 100 Hz over I2C or SPI, and supports on-chip detection of free fall, tap, double-tap, step count, and orientation. Each reading is a timestamped record of acceleration and angular velocity on all three axes; the sensor does not transmit autonomously, but responds to MCU polling or interrupt requests.

Health Sensor — Maxim MAX86176

The MAX86176 is a professional-grade optical biosensor selected as an upgrade path over the MAX30102 referenced in Section 6's original hardware table, integrating red and infrared LEDs with a photodiode detector in a 3.3×5.6 mm package. Optical heart-rate measurement works by emitting light into the skin and measuring the variation in reflected light caused by blood-volume changes with each heartbeat: LED emission, skin penetration, blood-volume change, light-reflection variation, and photodiode detection together produce a raw PPG waveform, which the MCU processes into heart rate, SpO_2 , HRV, and respiration rate — the direct inputs to the Health Anomaly AI (E3). The optical window must sit directly against the skin behind a light seal gasket, since ambient light leakage degrades measurement accuracy and no other sensor may sit between the skin and the PPG window.

GPS Module — u-blox MAX-M10S

The MAX-M10S provides multi-constellation GNSS positioning across GPS, GLONASS, Galileo, and BeiDou, reporting latitude/longitude, ground speed, heading, a position-fix accuracy estimate, and a UTC timestamp every few seconds (interval configurable for power management). The module only reports position; the GPS + Safe Zone AI (E10) and Digital Twin (E7) described in Section 6 are solely responsible for interpreting what that position means.

Environmental Sensors — Temperature, Barometer, Ambient Light

Component	Manufacturer	Function	Interface
TMP117	Texas Instruments	Skin/ambient temperature, $\pm 0.1^\circ\text{C}$ precision	I2C
BMP390	Bosch	Barometric pressure, altitude, floor detection	I2C
LTR329	AMS	Ambient light sensing	I2C

Microphone — Digital MEMS

A digital MEMS microphone (Knowles SPH0645 in the initial BOM, with INMP441 used as an equivalent part in later layout iterations) provides the I2S audio input to the Distress Audio AI (E4) and Safe Word / Panic Voice AI (E5) described in Section 6. It sits behind an acoustic membrane, such as a Gore Acoustic Vent, that preserves audio fidelity while maintaining water resistance.

Sensor-to-Engine Mapping

Sensor	Primary AI Engine(s)
BMI270 (motion)	E1/E1b Fall Detection, E2 Activity Recognition, E6 Attack Detection
MAX86176 (PPG / health)	E3 Health Anomaly AI, E6 Attack Detection (panic HR/HRV)
u-blox MAX-M10S (GPS)	E7 Digital Twin, E10 GPS + Safe Zone AI
TMP117 / BMP390 / LTR329	E3 Health Anomaly AI (context), E8 Context Engine
MEMS microphone	E4 Distress Audio AI, E5 Safe Word / Panic Voice AI

Table 23. Mapping of physical sensors to the AI engines (E1–E13) described in Section 6.

13.5 Communication Subsystem

Connectivity is provided by three physically separate radios, each with its own antenna, supporting the Phone-Only and Full Wearable operating modes defined in Section 12.3.

LTE Module — SIMCom SIM7080G

Technology	NB-IoT and LTE-M (Cat-M1)
Package size	24 × 24 mm module (16.1 × 10.1 mm compact variant)
Interface to MCU	UART + GPIO
Power	Low-power optimized for wearable use

The SIM7080G was selected over the SIM7600 for its significantly lower power draw. It handles cellular registration, data transmission, message reception, and server communication only; it does not run AI or read sensors, and is the physical channel behind the “Independent SOS Without Phone” capability of Full Wearable Mode (Section 12.3).

eSIM — STMicroelectronics ST4SIM

A 6 × 5 mm embedded SIM chip storing the IMSI, encryption keys, and carrier profile required for network authentication; without it the LTE module cannot register on any mobile network.

BLE and Antenna Design

Three separate antennas — a ceramic-chip BLE antenna near the nRF52840, a ceramic-patch GPS antenna, and a flexible-PCB LTE antenna — each require a minimum 10 mm clearance from other antennas, metal components, and the battery to minimize interference. Antenna tuning and RF performance verification should be performed by an RF engineer during the prototype phase.

13.6 Power Management

The Nordic nPM1300 power-management IC (5 × 5 mm) handles battery charging and protection, buck conversion, LDO regulation for sensitive analog circuits, fuel-gauge integration, and power sequencing for orderly boot-up and shutdown. A dedicated MAX17048 fuel-gauge IC (2 × 2 mm) provides precise state-of-charge monitoring.

Parameter	Value
Battery type	Li-Po (lithium polymer)
Capacity	600 mAh
Voltage	3.8V nominal
Dimensions	≈40 × 20 × 5 mm
Operating time	Up to 5–7 days normal use / up to 24 hours in emergency mode
Charge time	≈2.5 hours at 500 mA

Charging uses a 4-pin magnetic pogo-pin dock (gold-plated), following the same design philosophy as Garmin, Fitbit, and Huawei wearables, with a 5V input, an IP68-rated connector, and approximately 500–800 g of magnetic retention force.

Pin	Signal	Description
1	VBAT+	Battery positive (charging input)
2	GND	Ground
3	NTC	Battery temperature sense
4	CHG_EN	Charge enable / status

Table 24. Four-pin magnetic charging connector pinout.

13.7 Complete Bill of Materials

All twenty-one components are industrial-grade, selected for low power consumption, small size, and high reliability, and are available from established distributors (Mouser, DigiKey, LCSC).

#	Component	Part Number	Package	Qty
1	MCU	Nordic nRF52840	QFN73	1
2	Motion sensor	Bosch BMI270	LGA-14	1
3	Health sensor (PPG)	Maxim MAX86176	WLP/TQFN	1
4	Temperature sensor	Texas Instruments TMP117	SOT-563	1
5	Barometer	Bosch BMP390	LGA	1
6	Microphone	Knowles SPH0645	LCC	1
7	LTE module	SIMCom SIM7080G	Module	1
8	eSIM	STMicroelectronics ST4SIM	LGA	1
9	GPS module	u-blox MAX-M10S	QFN	1
10	Flash memory	Winbond W25Q128	SOP8/WSO8	1
11	Secure element	NXP SE050	TDFN	1
12	PMIC	Nordic nPM1300	QFN	1
13	Battery gauge	Maxim MAX17048	SOT-23-6	1
14	Battery	Li-Po 600 mAh	Custom	1
15	Vibration motor	Coin vibration motor	10 mm	1
16	Speaker	Micro speaker	10 × 15 mm	1
17	RGB LED	RGB LED	0603	1
18	Ambient light sensor	AMS LTR329	SMD	1
19	Touch controller	CAP1203	SOT-23-6	1
20	Charging connector	Magnetic 4-pin pogo	Custom	1
21	Antennas (BLE/GPS/LTE)	Ceramic + flex PCB	Various	3

Table 25. Complete bill of materials for the EVT prototype.

13.8 PCB Design and Layout

Dimensions	61.0 × 22.0 mm (main board; RF daughter-board variants run smaller)
Layers	4-layer
Total thickness	0.8–1.0 mm
Material	FR4 High TG (170°C)

Four-Layer Stackup

Layer	Content
L1 (Top)	Components and high-speed signals
L2	Solid ground plane
L3	Power plane (3.3V / 1.8V / VBAT)
L4 (Bottom)	Signals and low-speed traces

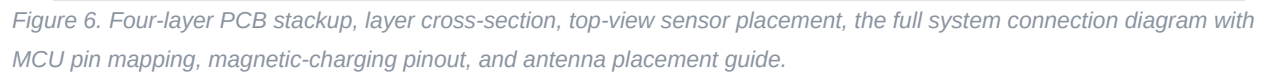
Component Interface Map

Component	Interface	Connection
BMI270	I2C or SPI	nRF52840
MAX86176	I2C	nRF52840
TMP117 / BMP390	I2C	nRF52840
LTR329	I2C	nRF52840
Microphone (I2S)	I2S	nRF52840
CAP1203 (touch)	I2C	nRF52840
W25Q128 (flash)	SPI	nRF52840
SE050 (secure element)	I2C	nRF52840
SIM7080G (LTE)	UART + GPIO	nRF52840
u-blox GPS	UART	nRF52840 (or via LTE module)
nPM1300 (PMIC)	I2C	nRF52840
MAX17048 (fuel gauge)	I2C	nRF52840
RGB LED	GPIO/PWM	nRF52840
Vibration motor	MOSFET + GPIO	nRF52840
Speaker	Audio amp + I2S/PWM	nRF52840
Magnetic charger	Direct	nPM1300

Table 26. Component interface map — every peripheral communicates with the nRF52840 MCU, consistent with the centralized-orchestration philosophy of Section 13.1.

PCB Layout Guidelines

- Separate LTE, GPS, and BLE antennas by at least 10 mm to reduce interference.
- Place the PMIC close to the battery to minimize power loss.
- Keep PPG sensor traces short and isolated from RF signals.
- Add test points for all power rails and interfaces (SWD, UART, I2C, SPI).
- Use ground stitching vias around RF sections.
- Follow Nordic and SIMCom reference designs for impedance matching.



The sketch below was produced during early ideation as a rough internal-architecture concept render. It is included here for design-process traceability only. It is not a verified engineering drawing: several dimensions and component labels shown differ from, or are illegible relative to, the specification fixed in Section 13.9 and the verified reference sheets in Figures 5, 6, 9, and 10.



13.9 Mechanical and Industrial Design

Length	70.2 mm
Width	27.0 mm
Thickness	15.5 mm
Weight	≈38 g with strap
Strap width	20.0 mm

Internal Assembly Stack

From the outer (top) cover down to the skin-contact side: top cover; LED light guide; top PCB (sensor board); sensor PCB; main PCB (MCU and system); bottom PCB; Li-Po battery (3.8V, 600 mAh); bottom case; and the 4-pin magnetic connector.

Materials Selection

Part	Material	Rationale
Outer shell	PC + ABS (polycarbonate / acrylonitrile butadiene styrene)	Strong, lightweight, impact-resistant, injection moldable, heat-resistant
Strap	Liquid Silicone Rubber (LSR)	Soft, hypoallergenic, flexible, water-resistant, easy to clean
Sensor window	Sapphire glass (preferred) or Gorilla Glass with AR coating	Optical clarity for PPG accuracy; anti-reflective coating reduces stray light
Waterproof gasket	Silicone O-ring / liquid silicone gasket	IP67/IP68 water resistance
Charging pins	Gold-plated pogo pins	Corrosion resistance, reliable contact

Waterproofing and Skin-Contact Sensor Stack

The band targets an IP68 rating (1.5 m depth for 30 minutes), achieved through a silicone gasket between the top and bottom covers, four stainless-steel M1.4/M1.6 screws for secure clamping, a snap-fit top cover, and an acoustic membrane (Gore Acoustic Vent) over the microphone. On the skin-contact side, a light-seal ring surrounds the PPG sensor to prevent ambient light leakage; no other sensor is placed between the skin and the PPG window.

<i>Skin → optical glass → light seal → MAX86176 PPG sensor → PCB → battery</i>
--

Assembly Notes

- The PCB is mounted on standoffs with M1.4/M1.6 screws.
- The battery is secured in a tray with a foam pad and Kapton tape rather than glued directly.
- The vibration motor is mounted in a silicone holder to isolate vibration from the PCB.
- The speaker requires a dedicated acoustic chamber.

13.10 Firmware Architecture

Firmware is partitioned into clearly separated layers rather than a single monolithic codebase, for maintainability and extensibility:

<i>Bootloader → Hardware Abstraction Layer (HAL) → Device Drivers → Sensor Manager → Power Manager → Communication Manager → AI Runtime → AI Models → Sensor Fusion Engine → Risk Engine → Emergency Orchestrator → Application Layer</i>	
Layer	Responsibility
Bootloader	Secure boot, firmware update handling
HAL	Abstracts hardware-specific details from higher layers
Device drivers	I2C, SPI, UART, GPIO, PWM, ADC drivers
Sensor manager	Configures and reads all sensors at scheduled intervals
Power manager	Manages sleep/wake cycles, battery monitoring, PMIC control
Communication manager	Selects optimal connectivity (BLE/LTE), manages connections
AI runtime	TinyML inference engine for on-device ML models
AI models	Motion classification, health analysis, fall-detection models (E1–E13)
Sensor fusion	Combines data from multiple sensors for context awareness
Risk engine	Calculates a risk score (0–100) from fused data (E9)
Emergency orchestrator	Coordinates emergency response actions
Application layer	User-facing logic, button handling, LED patterns, UI behavior

Table 27. Firmware software layers, from lowest (bootloader) to highest (application) level of abstraction.

User Interface Behavior

Action	Function
Short press	Confirm / acknowledge alert
Double tap	Dismiss / stand down alert
Long press (3s)	Emergency SOS
Long press (6s)	Silent Emergency Mode (see Engine E6, Section 6)

LED Color	Meaning
Green	Safe / normal operation
Blue	Connected (BLE/LTE active)
Orange / yellow	Warning / low battery
Red	Emergency / critical alert
Purple	Charging in progress

Table 28. Touch-button actions and RGB LED status codes.

13.11 On-Device AI Pipeline and Alignment with the Engine Architecture

The on-device AI pipeline runs entirely on the nRF52840 using TinyML, executing several specialized models in parallel: a Motion AI classifying activity (walking, running, sleeping, stationary), a Health AI analyzing heart-rate patterns, HRV, and SpO₂ trends, a Fall Detection AI identifying sudden-impact-plus-motion-cessation patterns, and a Context AI fusing motion, GPS, and health data for situational awareness. These four on-device models are the hardware-level realization of the “Band (edge)” rows of the Model Architecture tables given for Engines E1, E1b, E2, E3, and E8 in Section 6; the corresponding phone- and cloud-tier models described there run on the companion mobile application and backend rather than on the band itself.

The Risk Engine (E9) computes a continuous 0–100 risk score from the outputs of all on-device and phone-tier AI models; low scores indicate normal activity, while high scores trigger escalation. The AI layer never contacts the outside world directly: a high-risk determination is passed to the Emergency Orchestrator, which sits between the Risk Engine and the Communication Manager and is solely responsible for coordinating the appropriate response — the hardware/firmware counterpart of the autonomous escalation logic described in Section 5.

Worked Example — Fall Detection Flow

BMI270 → sudden impact detected → Fall AI → possible fall → Health AI → heart-rate spike confirmed → GPS → subject stopped moving → Risk Engine → risk score = 95 → Emergency Orchestrator → activate emergency state → LTE → cloud → guardian alert → mobile app notification → emergency services (if enabled in region)

This sequence is the hardware/firmware-level counterpart of the Engine E1 example in Section 6 and the Incident Summary timeline in Section 12.4: the same fall event is recorded once as a firmware decision trace and once as a human-readable incident record, from two layers of the same architecture.

13.12 Runtime Data Flow — A Typical One-Second Cycle

Consider a user who is walking outdoors. Within a single sensing cycle: the BMI270 sends motion data, which the MCU processes and the Motion AI classifies as “Walking”; the MAX86176 sends PPG data, from which the Health AI computes a heart rate of 72 BPM; the GPS module provides the current location; the Context AI fuses these into “Walking Outdoors”; the Risk Engine evaluates a risk score of 5 (low); and the dashboard (Section 12, Screen 11 or Screen 23) updates with the latest status. This cycle repeats continuously at a frequency governed by the Power Manager.

Channel	Protocol	Use Case
BLE	Bluetooth 5.3	Short-range connection to the mobile app; configuration; firmware updates
LTE-M / NB-IoT	Cellular	Long-range data transmission; emergency alerts; cloud sync

Table 29. Communication channels used during normal and emergency operation.

13.13 Development Roadmap and Team Composition

Building a production-quality prototype requires six specialized roles; no single person typically covers all of them.

Phase	Role	Responsibilities	Typical Tools
1	Industrial Designer	External form factor, ergonomics, button/LED placement, strap design, comfort	SolidWorks, Fusion 360, Rhino
2	Mechanical Engineer	Internal structure, snap-fits, screw locations, antenna zones, gaskets, battery tray	SolidWorks, Fusion 360
3	PCB Designer	Component placement, routing, impedance control, RF layout, power/ground planes	Altium, KiCad, OrCAD
4	RF Engineer	Antenna placement, keep-out zones, matching circuits, antenna tuning, RF compliance	HFSS, CST, VNA
5	DFM Engineer	Design-for-manufacturing review, assembly feasibility, cost optimization	—
6	Prototype House	PCB fabrication, 3D printing, assembly, initial testing	PCBWay, JLCPCB, Seeed Studio

Table 30. The six specialized roles typically required to take SENTRIX CORE from concept to a validated hardware prototype.

Recommended Prototype Path

Injection molding should not be the starting point, since production molds can cost thousands of dollars each. The recommended sequence is instead: (1) 3D-print the enclosure using SLA resin or SLS nylon for high-quality prototyping; (2) test the fit and assembly of all components; (3) iterate the design as needed; and (4) proceed to injection molding for mass production only after validation.

13.14 Key Specifications Summary

Table 31 consolidates the complete EVT prototype specification described throughout Section 10, targeting a design methodology consistent with commercial smart bands prior to mass production. All components were selected for availability, documentation quality, and industrial-grade reliability.

Category	Specification
MCU	Nordic nRF52840 (ARM Cortex-M4F, 64 MHz, 1 MB flash, 256 KB RAM)
Wireless	BLE 5.3 + LTE Cat-M1 / NB-IoT
Location	GPS / GLONASS / Galileo / BeiDou (u-blox MAX-M10S)
Motion	6-axis IMU (BMI270: accelerometer + gyroscope)
Health	PPG, heart rate, SpO ₂ , HRV, respiration (MAX86176)
Temperature	±0.1°C precision (TMP117)
Barometer	Pressure, altitude, floor detection (BMP390)
Microphone	Digital MEMS, I2S (Knowles SPH0645)
Ambient light	LTR329
Secure element	NXP SE050
Battery	Li-Po 3.8V, 600 mAh
Battery life	5–7 days normal / up to 24 hours emergency
Charging	4-pin magnetic pogo, 5V input, ≈2.5 hours
Waterproofing	IP68 (1.5 m, 30 minutes)
Dimensions	70.2 × 27.0 × 15.5 mm
Weight	≈38 g with strap
PCB	61.0 × 22.0 mm, 4-layer, FR4 High TG
Operating temperature	–20°C to 60°C
Storage temperature	–30°C to 70°C
Drop resistance	1.5 m
Body material	PC + ABS
Strap material	Liquid Silicone Rubber (LSR)
Sensor window	Sapphire glass or Gorilla Glass with AR coating

Table 31. Complete EVT prototype specification summary for the SENTRIX CORE smart safety band.

Taken together with the thirteen-engine AI architecture (Section 6), the seventy-four-screen mobile application (Section 12), and the multi-platform companion ecosystem (Section 12.7), this hardware specification completes the full-stack description of SENTRIX CORE: sensing hardware and industrial design, on-device and cloud AI, firmware, a consumer application, caregiver and institutional tooling, and emergency-responder integration.

14. Hardware Implementation Roadmap

Section 13 specified the SENTRIX CORE hardware as an engineering artifact — its processor, sensors, communication subsystem, power system, PCB, and mechanical design. This section specifies the complementary process artifact: the phased implementation plan by which that specification is converted into a working Engineering Validation Test (EVT) prototype. The plan is written for a team transitioning from a software-only background into hardware development, and is organized as eighteen sequential phases (Phase 0 through Phase 17), each with a defined goal, primary activities, and exit criteria.

14.1 Implementation Philosophy

Three principles govern the roadmap. First, a hardware product is developed by a multidisciplinary team — industrial design, mechanical engineering, PCB design, RF engineering, firmware, and design-for-manufacturing are distinct disciplines, and the roadmap is organized so that each can be engaged at the appropriate phase rather than all at once. Second, requirements are frozen early and deliberately: Phase 0 exists specifically to prevent the specification drift that is the most common cause of schedule overrun in hardware projects, since a change to a physical dimension or a component footprint after PCB layout has begun is far more costly than the same change made on paper. Third, the plan follows the same staged validation sequence used by established wearable manufacturers — evaluation boards, a breadboard proof of concept, a first prototype PCB, EVT, DVT, PVT, and mass production — rather than attempting to design a final production device on the first iteration.

Evaluation Boards (concept proof) → Prototype PCB v1 (core components operating) → EVT — Engineering Validation Test (first integrated prototype) → DVT — Design Validation Test (reliability and environmental robustness) → PVT — Production Validation Test (manufacturing readiness) → Mass Production

14.2 Phase-by-Phase Development Roadmap

Table 38 summarizes all eighteen phases; Section 13.3 then provides expanded detail for the phases most relevant to a team new to hardware development (Phases 0, 3, 4, and 11).

Phase	Name	Primary Goal
0	Freeze Requirements	Lock a single Hardware Specification v1.0 document (dimensions, components, AI features, battery-life target, IP rating, BOM) so that scope cannot silently drift during development.
1	Hardware Architecture	Define the system, power, and communication block diagrams (Section 13.2) before any component is purchased.
2	Final Component Selection	Lock the Bill of Materials (Table 25, Section 13.7); no further part substitutions without a documented reason.
3	Foundational Hardware Learning	Build working knowledge of voltage, current, power, buck/LDO regulation, I2C/SPI/UART/I2S/GPIO/ADC, PCB stackup, and RF basics.
4	Development Boards	Bring up every sensor and radio individually on its manufacturer evaluation board before any custom PCB is designed.
5	Firmware Bring-Up	Write and validate individual sensor drivers, then BLE, GPS, LTE, and an initial TinyML inference path, against the evaluation boards.
6	Power Optimization	Implement sleep modes, wake interrupts, sensor duty-cycling, LTE scheduling, and battery-life estimation against the Section 13.6 power budget.
7	PCB Schematic	Capture the full electrical schematic in Altium Designer or KiCad, matching the component interface map (Table 26, Section 13.8).
8	PCB Layout	Route the four-layer board (Section 13.8.2): component placement, high-speed and RF routing, ground and power planes, antenna keep-out zones.
9	Mechanical Design	Model the full enclosure in SolidWorks or Fusion 360, following the internal assembly stack defined in Section 13.9.
10	Prototype Fabrication	Produce an SLA-printed enclosure and a fabricated PCB, then hand-assemble the first physical unit.
11	Bring-Up Testing	Power on the assembled prototype for the first time and verify every subsystem individually before integrated testing.
12	AI Integration	Deploy the trained TinyML models (Fall Detection, Activity, Health, Risk Engine, Predictive AI) onto the on-device inference runtime (Section 13.11).
13	Mobile Integration	Connect the firmware's BLE GATT interface to the Flutter mobile application, cloud API, and Guardian ecosystem (Section 12).
14	EVT	Engineering Validation Test: verify that every specified function works under controlled conditions (fall detection, walking, running, heart rate, GPS, LTE, charging, battery life).
15	DVT	Design Validation Test: verify robustness under environmental stress (rain, heat, cold, perspiration exposure, drop, vibration) ahead of design freeze.
16	Certification	Obtain regulatory approval: FCC, CE, Bluetooth SIG qualification, RoHS, and UN38.3 (lithium battery transport).
17	Pilot Production	Scale assembly in controlled batches — 10, then 50, then 100, then 1,000 units — validating manufacturing repeatability before full-scale production.

Table 38. The eighteen-phase SENTRIX CORE hardware implementation roadmap, from requirements freeze (Phase 0) to pilot production (Phase 17).

14.3 Expanded Detail for Early-Stage Phases

Phase 0 — Freeze Requirements

Deliverable: a single controlled document, *SENTRIX CORE Hardware Specification v1.0*, covering final dimensions, the locked component list, the AI feature set the hardware must support, the battery-life target, the IP water-resistance rating, LTE/BLE/GPS requirements, the sensor suite, the system power budget, and the Bill of Materials. Once issued, this document may only be revised for a documented engineering reason, not general iteration — the objective is to prevent the specification churn that is the most common source of schedule slippage in a first hardware project.

Phase 3 — Foundational Hardware Learning

Before any schematic or PCB layout work begins, the team should establish working fluency in the following areas, since every subsequent phase assumes this vocabulary:

- Electrical fundamentals: voltage, current, and power relationships relevant to a battery-powered device.
- Power conversion: buck converters and LDO (low-dropout) linear regulators, and when each is appropriate.
- Serial communication buses: I2C, SPI, UART, and I2S, and which of the Section 13 sensors use each.
- General-purpose digital and analog I/O: GPIO and ADC.
- Multi-layer PCB stackup fundamentals (Section 13.8.1).
- RF basics: antenna keep-out zones, impedance matching, and the reasons the three SENTRIX CORE radios (Section 13.5) require physical separation.

Phase 4 — Development Boards and Evaluation Kits

Every sensor and radio in the Section 13 BOM is first brought up individually on its manufacturer evaluation board, wired to a general-purpose development kit, before any commitment is made to a custom PCB. This isolates sensor-level bring-up problems from PCB-level problems, and lets driver-level firmware be written and debugged against known-good reference hardware. Table 39 lists the recommended initial procurement set.

Item	Purpose
Nordic nRF52840 Development Kit	Primary MCU bring-up and BLE stack development
Bosch BMI270 evaluation board	Motion sensor driver development (Engines E1/E1b/E2)
MAX86176 evaluation kit (or MAX30102 as a lower-cost starting substitute)	PPG/health sensor driver development (Engine E3)
SIM7080G evaluation board	LTE modem bring-up and AT-command / data-session testing
u-blox MAX-M10S GPS module	GNSS fix acquisition and NMEA parsing
Li-Po battery, 3.7V, 500–600 mAh	Power-system bring-up and battery-gauge calibration
Nordic nPM1300 evaluation kit	PMIC and charging-path bring-up
Logic analyzer	Bus-level debugging of I2C/SPI/I2S traffic
USB oscilloscope (recommended)	Power-rail and signal-integrity verification
J-Link debugger (preferred with Nordic silicon)	Firmware flashing and hardware-level debugging

Table 39. Recommended initial evaluation-hardware procurement set for Phase 4.

Phase 11 — Bring-Up Testing

The first power-on of an assembled prototype is treated as a structured test, not a demo. Subsystems are verified individually and in this order, so that a failure in one subsystem does not obscure the diagnosis of another: battery and power rails, MCU boot, individual sensors (motion, PPG, temperature, barometer, microphone) over their respective buses, BLE advertising and pairing, GPS fix acquisition, LTE network registration, and finally the magnetic charging path.

14.4 Required Design Tools and Software

Discipline	Recommended Tools
PCB design	KiCad (open-source) or Altium Designer
Mechanical CAD	SolidWorks or Autodesk Fusion 360
Firmware IDE	SEGGER Embedded Studio, or Visual Studio Code with the nRF Connect SDK
Debugging	J-Link debugger paired with a logic analyzer
PCB fabrication and assembly	JLCPCB or PCBWay
Enclosure prototyping	SLA resin printing initially; SLS or injection molding for later production stages

Table 40. Design, fabrication, and validation tooling recommended across the implementation roadmap.

14.5 Component Footprint Reference (Actual Package Size)

Table 41 restates every major component from the Section 13 Bill of Materials with its exact package footprint, so that PCB floor-planning and enclosure clearance can be verified against real, to-scale dimensions rather than approximate sizing. All figures are manufacturer package dimensions, not including solder-mask or placement keep-out margin.

Component	Package	Footprint (L × W)	Height
Nordic nRF52840 (MCU)	QFN73	7.0 × 7.0 mm	0.9 mm
Bosch BMI270 (motion)	LGA-14	2.5 × 3.0 mm	0.8 mm
Maxim MAX86176 (PPG/health)	WLP/TQFN	3.3 × 5.6 mm	1.1 mm
Texas Instruments TMP117 (temperature)	SOT-563	2.0 × 2.0 mm	0.6 mm
Bosch BMP390 (barometer)	LGA	2.0 × 2.0 mm	0.8 mm
Knowles SPH0645 (microphone)	LCC	3.5 × 2.65 mm	0.9 mm
SIMCom SIM7080G (LTE, compact variant)	Module	16.1 × 10.1 mm	2.0 mm
u-blox MAX-M10S (GPS)	QFN	9.7 × 10.1 mm	2.4 mm
STMicroelectronics ST4SIM (eSIM)	LGA	5.0 × 6.0 mm	0.7 mm
Winbond W25Q128 (flash)	SOP8 / WSON	5.0 × 6.0 mm	0.8 mm
NXP SE050 (secure element)	TDFN	2.5 × 2.5 mm	0.6 mm
Nordic nPM1300 (PMIC)	QFN	5.0 × 5.0 mm	0.9 mm
Maxim MAX17048 (fuel gauge)	SOT-23-6	2.0 × 2.0 mm	1.0 mm
Coin vibration motor	Cylindrical	10.0 mm diameter	2.6 mm
Micro speaker	Cylindrical	15.0 mm diameter	3.4 mm

Table 41. Component footprint reference at actual package size, consolidated from manufacturer datasheets for PCB floor-planning against the 61.0 × 22.0 mm board outline specified in Section 13.8.

14.6 Wristband Removal and Tamper Detection

A safety wearable must be able to distinguish an ordinary, voluntary removal of the band from a forced or tampered removal, since the latter is itself a safety-relevant event (for example, an attacker removing the band from a Women Safety Mode user to disable tracking). SENTRIX CORE addresses this with a two-layer design: a lightweight, rule-based hardware detection layer that determines only whether the band is being worn, and an AI classification layer, implemented as a submodule of the existing Motion AI Engine, that determines how the band came off.

Layer 1 — Hardware Wear Detection (Rule-Based)

Wear state is determined directly from existing sensors, with no machine-learning inference required: the capacitive touch/skin-contact sensing already used for the logo button, continuity of the PPG signal from the MAX86176 (Section 13.4), and physical strap-closure state are combined into a simple rule.

Skin contact lost + no PPG signal + no motion for a sustained interval → Band Removed

This rule alone is sufficient for the large majority of commercial wearables and requires no AI model; it answers only “is the band currently on the wrist?” and does not attempt to characterize how it came off.

Layer 2 — Forceful Removal Detection AI (Motion AI Submodule)

Determining *how* the band was removed requires interpreting the motion signature leading up to the removal event, which is why this capability is implemented as a submodule of the Motion AI Engine (Section 6, Engines E1/E1b/E2) rather than as a new, independent top-level engine — it consumes the same accelerometer and gyroscope streams already processed by fall and activity detection, and reuses the same on-device inference path.

Input	Source
Accelerometer	BMI270 (Section 13.4)
Gyroscope	BMI270 (Section 13.4)
Skin-contact state	Capacitive touch / PPG continuity
Heart rate	MAX86176 (Engine E3)
Recent motion history	Motion AI Engine buffer

The classifier distinguishes four outcomes. A **Normal Removal** shows calm, gradual motion, a stable heart rate, and a strap that opens smoothly — no action is taken beyond logging. An **Accidental Slip** shows a brief, low-energy disturbance with no sustained struggle pattern. A **Forceful Removal** shows a high-magnitude acceleration spike, a sudden, fast wrist rotation, an abrupt loss of skin contact, and frequently an elevated heart rate immediately beforehand. A **Tampering Attempt** shows repeated partial-removal motion consistent with an attempt to disable the device without fully removing it. Table 42 summarizes the resulting system response.

Classification	System Response
Normal Removal	Log the event; no alert
Accidental Slip	Log the event; no alert unless it recurs frequently
Forceful Removal	Raise the Risk Score (Section 7); log the event with sensor evidence; notify the Guardian; escalate further if corroborated by other risk indicators (Section 7.5)
Tampering Attempt	Log the event with sensor evidence; notify the Guardian; treat repeated attempts as an elevated-risk pattern

Table 42. Band removal classifications and the corresponding system response.

Placement Within the Motion AI Engine

Motion AI Engine

- *Fall Detection AI (E1)*
- *Fall Prediction AI (E1b)*
- *Activity Recognition AI (E2)*
- *Gesture Detection AI*
- *Tremor Detection AI*
- *Wear Detection Module (rule-based, Layer 1)*
- *Forceful Removal Detection AI (Layer 2, this section)*

This placement keeps the system's AI inventory simple and avoids training and maintaining a fully independent model for a task that shares its entire input space with existing motion engines. A fully independent, separately trained Forceful Removal model would only be justified if a sufficiently large dedicated dataset (on the order of thousands to millions of labeled removal events) became available to support standalone research publication or a materially different model architecture — not a requirement for the current EVT-stage design.

14.7 Industrial Design Reference

Figures 9 and 10 present two converging industrial-design reference sheets used to finalize the SENTRIX CORE exterior identity described mechanically in Section 13.9: a curved, WHOOP-inspired band silhouette carrying a distinct SENTRIX identity, in which the logo itself functions as the capacitive touch control (Section 13.10.2). Both sheets are shown for traceability, since each contributed different elements to the final design — exterior dimensions and an exploded external view in Figure 9, and materials, on-wrist ergonomics, and colorway options in Figure 10.



Figure 9. Industrial design reference sheet A: exterior views and dimensions, exploded external assembly, skin-contact sensor layout, magnetic charging port detail, and size comparison against WHOOP 4.0 and Garmin Vivosmart 5.



Figure 10. Industrial design reference sheet B: perspective render, materials and finish callouts, feature highlights, and the three proposed colorway variants (Stealth Black, Deep Blue, Arctic Grey).

Both sheets converge on the same core dimensional envelope established in Section 13.9 (70.2 × 27.0 × 15.5 mm, approximately 38 g with strap), confirming that the industrial-design exploration remained consistent with the mechanical and PCB constraints fixed earlier in the specification rather than requiring them to be renegotiated.

15. Conclusion

The literature reviewed in this paper demonstrates substantial progress across wearable healthcare, fall detection, remote patient monitoring, edge AI, and Digital Twin modeling considered individually. However, both the academic literature and the commercial product landscape remain fragmented: each system addresses one facet of personal safety — health, location, emergency response, or personalization — without integrating them into a single, autonomous decision-making platform.

SENTRIX CORE addresses this gap through a thirteen-engine AI architecture that unifies smart wearable hardware, edge AI inference, a personalized Digital Twin, multi-modal sensor fusion, predictive risk assessment, and autonomous emergency response within a single coherent system. This architecture is delivered to end users through the seventy-four-screen application taxonomy described in Section 12, spanning core onboarding, daily health and tracking use, specialized modes for children, elderly users, and women's safety, a complete emergency and Guardian subsystem, and a wider multi-platform companion ecosystem, and it is realized in physical hardware through the EVT-grade smart-band specification in Section 13 — a nine-subsystem, MCU-orchestrated device with an industrial-grade sensor suite, LTE/BLE/GPS connectivity, and an on-device TinyML pipeline that together supply the raw signals and edge inference the AI architecture depends on. Section 8 completed the emergency-response picture with a full specification of the Nearby Rescue Network — the community-response layer that, together with the Guardian ecosystem and professional emergency services, embodies the “predict and coordinate, not just alert” positioning argued for in Section 3.1 — and Section 14 translated the hardware specification into an eighteen-phase implementation roadmap, from requirements freeze through pilot production. SENTRIX CORE is therefore proposed not as a replacement for existing wearable devices, but as a next-generation safety-intelligence platform — spanning silicon, firmware, application, and cloud — that can be layered on top of, or integrated with, the wearable hardware category as a whole.

Future work should focus on empirical validation of the fused Risk Score Engine (E9) against confirmed real-world emergency outcomes, prospective clinical evaluation of the Health Anomaly and Attack/Kidnapping Detection engines, a formal patentability assessment of the three sub-systems identified in Section 10, usability testing of the application's Phone-Only and Full Wearable operating modes described in Section 12, RF, thermal, and battery-life validation of the EVT hardware prototype described in Section 13 ahead of design-for-manufacturing review, and a field trial of the Nearby Rescue Network's helper-selection model (Section 8.4) to validate its estimated-time-to-arrival predictions against real responder behavior.

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